



User trust in artificial intelligence: A comprehensive conceptual framework

Rongbin Yang¹ · Santoso Wibowo^{1,2}

Received: 11 March 2022 / Accepted: 7 September 2022 / Published online: 4 November 2022
© The Author(s), under exclusive licence to Institute of Applied Informatics at University of Leipzig 2022

Abstract

This paper provides a systematic literature review of current studies between January 2015 and January 2022 on user trust in artificial intelligence (AI) that has been conducted from different perspectives. Such a review and analysis leads to the identification of the various components, influencing factors, and outcomes of users' trust in AI. Based on the findings, a comprehensive conceptual framework is proposed for a better understanding of users' trust in AI. This framework can further be tested and validated in various contexts for enhancing our knowledge of users' trust in AI. This study also provides potential future research avenues. From a practical perspective, it helps AI-supported service providers comprehend the concept of user trust from different perspectives. The findings highlight the importance of building trust based on different facets to facilitate positive cognitive, affective, and behavioral changes among the users.

Keywords AI · Trust · User · Literature review · Comprehensive framework

JEL classification M15

Introduction

The application of artificial intelligence (AI) has presented new opportunities for organizations to improve their performance through improved decision making. Since 2015, a few countries have released national-level AI strategic plans (Fatima et al., 2021); thus, the number of AI-based applications and related studies in different fields has increased dramatically in recent years (Baum et al., 2021; Fatima et al., 2021; Paek & Kim, 2021; Pena et al., 2021). Sutrop (2019) states that AI is now increasingly used in diverse organizational practices, creating new types of human-machine configurations and playing an increasing role in contemporary organizing. Wei et al. (2019)

argue that AI enables organizations to analyze customer data and predict what products the customers are likely to purchase in the near future. Meanwhile, Jones (2018) points out that AI takes this data analysis one step ahead. For example, Apple's Siri is an intelligent software assistant that can interpret voice commands, pick up relevant keywords, and execute a set of built-in commands and responses. This software is capable of accessing personal information stored on a user's phone, as well as predicting and responding to the user's requests and inquiries.

The adoption of AI, however, has led to various concerns, such as consumer trust, interactions, and perceived risk, which have a direct influence on effective relationship management between consumers and organizations (Hsiao & Chen, 2021). Hasan et al. (2021) find that consumers often feel conflicted on whether they should adopt new technologies due to trust, ethics, data collection, usage, and privacy issues, especially after the Facebook data breach in 2018. This incident involved more than 50 million users' data harvested for third parties, and it has significantly impacted people's trust in online platforms used for mostly non-commercial activities (Hasan et al., 2021; Landwehr, 2019; Liyanaarachchi et al., 2020). Yen & Chiang (2021) believe that even though the use of AI is on the rise, organizations need to focus on building trust with consumers and assure them that their privacy and personal information

Responsible Editor: Veda Storey

✉ Rongbin Yang
Rongbin.yang@cquemail.com

¹ Kaplan Business School Adelaide Campus, 1/68 Grenfell St, Adelaide, SA 5000, Australia

² Central Queensland University, 120 Spencer St, Melbourne, VIC 3000, Australia

will not be compromised. Thus, it is critical for organizations to understand the impact of AI adoption on trust and their business practices.

While user trust and the adoption of AI had become an emerging research topic in earlier studies prior to 2015, they tended to focus on the effects of technological factors, e.g. accuracy (Stumpf et al., 2007), security (Liao et al., 2011), explainability (Pieters, 2011), creativity (Skulimowski, 2013), collaboratively (Li et al., 2014), design (Luxton, 2014). Despite the increase in empirical research attempts to understand the adoption of AI and its relation to trust and organizational practices, systematic reviews of the existing studies on this topic from various perspectives are still lacking. The growing popularity of AI applications makes a systematic review of the related literature critical for summarizing what has been done in this area and, at the same time, establishing a solid foundation for future research in the domain. Therefore, the present study is aimed at addressing this gap with the formulation of the following research questions:

- *What are the components of trust in relation to AI?*
- *What are the critical factors that influence users' trust in AI?*
- *What are the outcomes of users' trust in AI?*
- *What is an appropriate conceptual model for better understanding users' trust in the adoption of AI?*

To address these research questions, a systematic review of the related literature on user trust in AI was conducted. A total of 131 papers published between January 2015 and January 2022 were synthesized to identify the commonly occurring themes. This enabled the identification of relevant factors related to user trust that are affecting the adoption of AI in practice. Subsequently, an integrated framework was developed to better tackle these issues, which can further be tested and validated to enhance the knowledge of user trust and the adoption of AI.

This paper is divided into seven sections. "[Research method](#)" of this paper presents the methodology used for this study. "[Overview of AI and user trust](#)" presents an overview of the development of AI and user trust. "[Literature review on trust in AI](#)" provides a review of related literature. "[Relevant theories on new technology adoption](#)" provides the discussions and a comprehensive conceptual framework that illustrates the components, influencing factors, and outcomes of user trust in AI. "[The components of trust in relation to AI](#)" presents the limitations and future research. "[Predictability, dependability, faith](#)" concludes the paper.

Research method

The aim of this research study was to explore the issues related to user trust in AI based on the literature from the past years and thereby guide future research in this area. To this end,

Wolfswinkel et al.'s (2013) five-phase approach was adopted for conducting a systematic review and analysis of the literature. This approach enables researchers to conduct a thorough search process and critically review and analyze the articles retrieved from various databases. The five-phase approach includes (a) defining the scope of the review, (b) searching the literature, (c) selecting the final samples, (d) analyzing the samples using content analysis, and (e) presenting the findings.

Accordingly, the first phase of the present study was to define the scope of the review, which included defining the criteria for the inclusion and exclusion of relevant sources and the criteria for identifying and retrieving those sources in the literature. Several databases, including Web of Science, ScienceDirect, and Emerald, were used due to their reputation, representativeness, and coverage in publishing top academic papers in relevant fields (Zhu & Liu, 2020). For a broad coverage of the studies obtained from these databases, the following keywords were used for the search: artificial intelligence, AI, customer, user, and trust. The criteria for the inclusion and exclusion of articles were based on our research objectives. We included articles published from January 2015 to January 2022 because the volume of AI-related applications and research in different fields has dramatically increased, especially after 2015 (Baum et al., 2021; Fatima et al., 2021; Paek & Kim, 2021; Pena et al., 2021), and the search in databases was conducted in early February 2022.

The second phase involved in running the search query within the selected databases and retrieving the search results. A total of 2559 articles were returned using the above pre-defined search strings. Then, 810 duplicated articles were removed, and a total of 1749 articles remained for further assessment. This initial search enabled us to gain a general understanding of the coverage of user trust in AI-related research.

The third phase required selecting the final samples for a detailed analysis. We applied the following criteria for the inclusion and exclusion of articles based on our research objectives, after the initial screening. First, we only included full-text articles published in academic journals and peer-reviewed conference proceedings related to our research objectives. Therefore, 895 publications (e.g. abstract-only or research-in-progress papers) were excluded. Second, studies presented in languages other than English were not considered to minimize regional over-representation of research in the formal analysis (Dreyer et al., 2019). Third, we excluded articles without a strong relevance to user trust, AI and its applications. Thus, the titles and abstracts of all the initial articles were screened for their relevance to our research topic. Accordingly, another 723 papers were removed. Then, we conducted a backward and a forward snowballing research recommended by Webster and Watson (2002). Accordingly, we adopted Google Scholar to find relevant publications from the references of the identified articles, resulting in 11 additional articles. After the above process, a total of 142 articles remained for further analysis.

Table 1 The Wolfswinkel et al.'s (2013) five-phase approach

Phases	Tasks	Outcomes
1. Define	Defining the scope of the review	Databases: Web of Science, ScienceDirect, and Emerald Keywords used for the search: artificial intelligence, AI, customer, user, trust. Time range: articles published from January 2015 to January 2022
2. Search	Searching the literature	2568 articles were returned using the pre-defined search strings. A total of 1758 articles remained for further assessment after removing 810 duplicated articles.
3. Select	Selecting the final samples	A total of 142 articles meeting the criteria remained for further analysis
4. Analyze	Analyzing the samples using content analysis	Articles were analyzed based on the proposed research questions related to the components of trust in relation to AI, critical factors influencing users' trust in AI, and the consequences of users' trust in AI.
5. Present	Presenting the findings	Table 2 and Table 3 were presented to summarize the key findings. A conceptual model was developed and presented based on the literature review (Fig. 1).

In the fourth phase, the 142 articles were analyzed based on the proposed four research questions. First, this research explored the different components of trust in relation to AI. Second, we investigated the critical factors influencing users' trust in AI. Third, various outcomes of users' trust in AI were identified. Finally, we summarized the relevant studies and grouped them into the above-mentioned categories as the foundation for a new conceptual framework.

For the last phase, several tables were presented to summarize the key findings, including the relevant theories in the field, components of trust, influencing factors and outcomes of users' trust. Based on the key findings, we developed a conceptual framework based on the literature review and presented it as a figure. Table 1 summarizes how this study was conducted based on Wolfswinkel et al.'s (2013) five-phase approach.

Overview of AI and user trust

AI has increasingly gained attention due to its capabilities in supporting common applications in everyday life, such as chatbots, facial recognition, image recognition, personalized recommendations, and other services (Wei et al., 2019; Hsiao & Chen, 2021). Even though AI positively impacts how individuals and organizations interact, there is a growing focus on its trustworthiness. Hasan et al. (2021) believe that trust plays a key role in helping individuals accept and use new technologies. For example, trust can alleviate users' concerns about purchase transactions and product quality in electronic markets. Similarly, Yang & Wibowo (2020) find that trust plays an important role in users' acceptance of e-government. These studies demonstrate that trust is an important factor in helping users overcome their doubts and fears, resulting in the adoption of new technologies such as AI.

Several studies (Oliveira et al., 2017; Nordheim et al., 2019; Wei et al., 2019; Gillath et al., 2021) have been conducted on user trust in the context of AI. Oliveira et al.

(2017) find that customers who trust the interaction with AI robots are likely to choose services provided by these robots in the first place. Trust is especially crucial in contexts where traditional face-to-face businesses are conducted over virtual media, such as e-commerce and electronic and online banking (Ahmadi & Tutschku, 2017). Nordheim et al. (2019) state that trust plays an important role in users' acceptance and use of AI-enabled systems. This is because trust reduces the uncertainties and perceived risks in adopting new technologies. Wei et al. (2019) argue that trust can influence users' behavioral intentions in using AI-based systems. Their study shows that the greater the trust customers have in the system that the organization offers, the higher their positive mindset towards using the organization's services. Gillath et al. (2021) examine how the human-to-human attachment process can be applied to increase users' trust in AI. They reveal that reducing anxiety and enhancing security associated with AI helps to increase trust in AI. Therefore, it is important to look into the issues and challenges related to users' trust in the adoption of AI in organizations.

Literature review on trust in AI

Relevant theories on new technology adoption

Previous studies have shown that trust plays a vital role in increasing the adoption and diffusion of AI applications in different areas (Ahmadi & Tutschku, 2017; Gillath et al., 2021). There were various theoretical models developed by earlier studies on individuals' adoption of new technologies, including the Theory of Reasoned Action (TRA) (Fishbein & Ajzen, 1977), Theory of Planned Behavior (TPB) (Ajzen, 1991), Technology Acceptance Model (TAM) (Davis, 1985), Technology–Organization–Environment framework (TOE) (Tornatzky et al., 1990), Diffusion of Innovation (DOI) (Rogers, 2003), Unified Theory of Acceptance and Use of Technology (UTAUT) (Venkatesh

et al., 2003). These theories have remained dominant in the technology adoption literature in recent years (Fan et al., 2020; Giampietri et al., 2018; Hampshire, 2017; Hasan et al., 2021; Nasirian et al., 2017; Vasiljeva et al., 2021). Of the above, recent studies in the field have frequently adopted the UTAUT model to explore and test the relevant factors affecting individuals' adoption of new technologies (Ashfaq et al., 2020; Mostafa & Kasamani, 2021; Mou et al., 2017; Moussawi et al., 2021; Wibowo and Mubarak, 2020). These factors include performance expectancy, effort expectancy, social factors, and facilitating conditions. Moreover, some researchers have adopted the extended UTAUT framework (i.e. UTAUT2), which also includes users' hedonic motivation, perceived price value, and habits (Gansser & Reich, 2021). However, the original UTAUT and UTAUT2 models have been criticized for their exclusion of certain relevant factors, including trust (Hasan et al., 2021). Thus, recent studies have added the concept of trust to this model in their attempts to better explain AI adoption behaviors (Ashfaq et al., 2020; Mostafa & Kasamani, 2021; Moussawi et al., 2021).

In the recent years, the Corporate Digital Responsibility (CDR) framework has become an emerging focus in this field, which is "defined as the set of shared values and norms guiding an organization's operations with respect to the creation and operation of digital technology and data." (Lobschat et al., 2021, p. 876). This framework illustrates the CDR-building process and components, the influences on CDR, and the outcomes of CDR. Akin to the CSR theory, CDR emphasizes the values and norms in terms of organizations' responsibility for their stakeholders (e.g. investors, employees, customers, and suppliers) and the significant role of corporate decision-makers (Elliott et al., 2021; Liyanaarachchi et al., 2020; Lobschat et al., 2021; Weber-Lewerenz, 2021). However, CDR has distinctiveness in terms of its primary focus on the digital aspect due to the "exponential growth in technological development, malleability of technologies and data in use, and pervasiveness of technology and data" (Lobschat et al., 2021, p. 876). This suggests that businesses can facilitate customers' trust in AI by improving digital responsibility, such as ethical data management (Jelovac et al., 2021). While the CDR framework covers different stakeholder groups, there is still a need for more studies focusing on specific aspects (e.g. users) to complement it with a more in-depth understanding, such as behavior change models (Ryan, 2020; Lobschat et al., 2021). Table 2 provides an overview of the fundamental theories and their key constructs.

The components of trust in relation to AI

The concept of trust can be understood in general by studying an individual's overall trusting beliefs in other people and things. However, a person's specific trusting beliefs can

be inconsistent in different circumstances (Thiebes et al., 2021). Trust has previously been studied as an important factor that affects people's acceptance of new technologies (Ashfaq et al., 2020; Mostafa & Kasamani, 2021; Mou et al., 2017; Moussawi et al., 2021; Zhang et al., 2021). Notably, trust was often a general concept in those studies (Moussawi et al., 2021; Zhang et al., 2021). However, as a fundamental concept, trust has been widely researched in various social science disciplines such as sociology, social psychology, philosophy, management, and economics (Rheu et al., 2021). Thus, the components of trust in relation to AI should be studied from different perspectives.

Predictability, dependability, faith

Trust is a dynamic concept that involves multiple stages (Rheu et al., 2021). According to one of the most popular approaches to defining trust in the social sciences, this concept includes three major stages such as predictability, dependability, and faith (Rheu et al., 2021; Sheridan, 2019). Rheu et al. (2021) consider predictability as the consistency of a trustee's behaviors or performance in the long term. When an AI application's performance is consistent over an extended period of time and across various circumstances, it supports a more precise prediction of the application's action with a certain level of confidence (Nordheim et al., 2019). Therefore, predictability is related to understandability, where users will be able to predict an AI's future actions if they can understand how it works. This will encourage users to form a prevalent belief that the new technology is dependable (i.e. dependability) (Sheridan, 2019). Thus, a higher level of predictability indicates a lower likelihood of unintended consequences that are unanticipated and may pose negative impacts on society (Pringle et al., 2016). Faith is usually considered the final stage of the formation of trust. Only when an AI application's performance is believed to be predictable and dependable will the user establish their faith and willingness to rely on the technology when needed (Rheu et al., 2021; Sheridan, 2019).

Ability, benevolence, and integrity

Another most well-cited approach used to define trust in interpersonal relationships is to assess this concept based on multiple criteria, which include users' belief in AI's ability, benevolence, and integrity (Bedué & Fritzsche, 2021; Pickering, 2021; Rheu et al., 2021; Schmidt et al., 2020; Thiebes et al., 2021). Ability concerns the skill of an AI application in completing a task assigned by the user (Rheu et al., 2021; Thiebes et al., 2021), as well as whether the AI can correctly decide and perform the most important of multiple tasks (Sheridan, 2019). Several researchers, such as Gulati et al. (2019), Nundy et al.

Table 2 Fundamental theories and key constructs

Theories/ Theoretical frameworks	Critical factors	References
Theory of Reasoned Action	Behavioral Beliefs, Evaluation of Behavioral Outcomes, Normative Beliefs, Motivation to Comply, Attitude toward Behavior, Subject Norm, Behavioral Intention, Behavior	Fan et al. (2020)
Theory of Planned Behavior	Behavioral Beliefs, Evaluation of Behavioral Outcomes, Normative Beliefs, Motivation to Comply, Attitude toward Behavior, Subject Norm, Behavioral Intention, Behavior, Control Beliefs, Perceived Power, Perceived Behavioral Control	Giampietri et al. (2018)
Technology Acceptance Model	Perceived Usefulness, Perceived Ease of Use, Attitude toward Using, Behavioral Intention to Use, Actual Use	Hampshire (2017)
Technology–Organization–Environment framework	External Task Environment (e.g. industry characteristics and market structure, technology support infrastructure, government regulation), Organization (formal and informal linking structures, communication processes, size, slack), Technology (availability and characteristics), Technological Innovation, Decision Making	Vasiljeva et al. (2021)
Diffusion of Innovation	Relative Advantage, Compatibility, Complexity, Trialability, Observability, Adoption	Nasirian et al. (2017)
Unified Theory of Acceptance and Use of Technology	Performance Expectancy, Effort Expectancy, Social Factors, Facilitating Conditions, Behavioral Intention and Use Behavior	Hasan et al. (2021)
Unified Theory of Acceptance and Use of Technology 2	Performance Expectancy, Effort Expectancy, Social Factors, Facilitating Conditions, Behavioral Intention, Use Behavior, Hedonic Motivation, Price Value, Habit	Gansser & Reich (2021)
Corporate Digital Responsibility	Influencing factors (i.e. social and organizational contexts), CDR culture (i.e. shared values, specific norms, artifacts and CDR-related behaviors), outcomes (i.e. organizations, individual actors, institutional actors/society at large, artificial/technological actors)	Lobschat et al. (2021)

(2019), and Pickering (2021), also use the term “competency” to describe an AI’s ability based on its mastery of relevant knowledge. Robustness is also an akin concept that refers to the ability and flexibility to perform variations of the task (Sheridan, 2019). Benevolence measures the degree of alignment of the AI’s intention with that of the user (Gulati et al., 2019; Rheu et al., 2021; Thiebes et al., 2021). Alternatively, researchers sometimes use the term “motive” to describe benevolence based on whether a trustee acts solely in the trustor’s interests (Nundy et al., 2019). Integrity concerns whether the AI’s behaviors consistently follow the moral principles accepted by the user groups. Thus, the formation of trust goes beyond the predictability of an AI application with this component, requiring the AI’s intention to align with the user’s and the avoidance of egotism (Rheu et al., 2021).

Trust in performance, process, and purpose

The trust framework for IT artifacts is another model adopted by some recent studies, and it consists of the “performance, process, and purpose” dimensions (Meske & Bunde, 2020; Park, 2020; Schmidt et al., 2020). Notably,

this framework can be seen as a synthesis of the above-discussed predictability–dependability–faith model and ability–benevolence–integrity model. The performance aspect concerns the ability of an IT artifact to be responsible and accurate with information (Qin et al., 2020). It also considers the reliability of the IT artifact by considering the user’s perception of whether it is dependable to consistently complete the given tasks (Meske & Bunde, 2020; Park, 2020; Schmidt et al., 2020). Likewise, the process aspect concerns the predictability of the IT artifact and considers the integrity elements (i.e. data integrity, confidentiality, and user authenticity). Last, the purpose aspect includes not only designer benevolence and authorized data usage but also users’ faith and future reliance on the IT artifact (Ameen et al., 2021; Meske & Bunde, 2020; Park, 2020; Schmidt et al., 2020).

Trust in the technology and trust in the organization

Some researchers explore the trustees from different perspectives by studying AI applications as technology-supported services provided by different organizations (Söllner et al., 2016). Accordingly, the concept of trust in AI can be

Table 3 Key findings from the literature review

Perspectives		Critical factors	References
Components of Trust		Predictability	Nordheim et al. (2019); Rheu et al. (2021); Sheridan (2019)
		Dependability	Rheu et al. (2021); Sheridan (2019)
		Faith	Rheu et al. (2021); Sheridan (2019)
		Ability	Bedué & Fritzsche (2021); Gulati et al. (2019); Nundy et al. (2019); Pickering, (2021); Rheu et al. (2021); Schmidt et al. (2020); Sheridan (2019); Thiebes et al. (2021)
		Benevolence	Bedué & Fritzsche (2021); Gulati et al. (2019); Pickering (2021); Rheu et al. (2021); Schmidt et al. (2020); Thiebes et al. (2021)
		Integrity	Bedué & Fritzsche (2021); Pickering (2021); Rheu et al. (2021); Schmidt et al. (2020); Thiebes et al. (2021)
		Trust in Performance	Meske & Bunde (2020); Park (2020); Qin et al. (2020); Schmidt et al. (2020)
		Trust in Process	Meske & Bunde (2020); Park (2020); Schmidt et al. (2020)
		Trust in Purpose	Ameen et al. (2021); Meske & Bunde (2020); Park (2020); Schmidt et al. (2020)
		Trust in the Technology	Ahmadi & Tutschku (2017); Gillath et al. (2021); Łapińska et al. (2021); Li et al. (2021); Mal et al. (2018); Mou et al. (2017); Thiebes et al. (2021); Wei et al. (2019)
		Trust in the Organization	Ahmadi & Tutschku (2017); Gillath et al. (2021); Łapińska et al. (2021); Li et al. (2021); Mal et al. (2018); Mou et al. (2017); Thiebes et al. (2021); Wei et al. (2019)
		Cognitive Trust	Gillath et al. (2021); Glikson & Woolley (2020); Khaksari & Keyvanpour (2019); Kraus et al. (2021)
		Affective Trust	Gillath et al. (2021); Glikson & Woolley, (2020); Khaksari & Keyvanpour (2019); Kraus et al. (2021); Shi et al. (2021)
		Behavioral Trust	Khaksari & Keyvanpour (2019); Li et al. (2021)
		Trust in specific applications	Park (2020),; Qin et al. (2020)
Factors Influencing Trust in AI	Technology-related Factors	Service Quality	Ameen et al. (2021); Ashfaq et al. (2020); Balakrishnan & Dwivedi (2021); Bedué & Fritzsche (2021); Hsiao & Chen (2021); Kerasidou (2020); Kim & Peterson (2017); Lv et al. (2022); Nasirian et al. (2017); Nilashi et al. (2015); Oliveira et al. (2017); Sheridan (2019)
		Information Quality	Ahmadi & Tutschku (2017); Ashfaq et al. (2020); Bedué & Fritzsche (2021); Hong & Williams (2019); Kim et al. (2021); Kim & Peterson (2017); Nasirian et al. (2017); Nilashi et al. (2015); Thiebes et al. (2021); Yen & Chiang (2021)
		System Quality	Ahmadi & Tutschku (2017); Anwar (2020); Bedué & Fritzsche (2021); Chatterjee & Sreenivasulu (2021); Felzmann et al. (2019); Girsang et al. (2020); Hepworth et al. (2020); Kim & Peterson (2017); Lazanyi (2018); Nasirian et al. (2017); Nassar et al. (2020); Nilashi et al. (2015); Qin et al. (2020); Robinson (2020); Shin (2021); Triberti et al. (2020)
		Design Quality	Ameen et al. (2021); Anwar (2020); Faisal et al. (2016); Følstad et al. (2018); Hsiao & Chen (2021); Kim & Peterson (2017); Moussawi et al. (2021); Nilashi et al. (2015); Qin et al. (2020); Shi et al. (2021)
		Anthropomorphism	Bernotat et al. (2017); Følstad et al. (2018); Gulati et al. (2019); Hsiao & Chen (2021); Kaplan et al. (2021); Kim & Song (2021); Lazanyi (2018); Lee & Chen (2022); Li et al. (2021); Lu et al. (2022); Mohanty (2020); Moussawi et al. (2021); Pelau et al. (2021); Shi et al. (2021); Troshani et al. (2020); Yen & Chiang (2021)
	Organizational Factors	Image and Reputation	Burda & Teuteberg (2014); Hamid et al. (2015); Kaplan et al. (2021); Khaksari & Keyvanpour (2019); Oliveira et al. (2017); Qin et al. (2020); Nguyen & Pervan (2020); Nordheim et al. (2019); Ryan (2020); Tuteja et al. (2016); Xin et al. (2015)
		Organizational Policy and Culture	Jain et al. (2020); Janssen et al. (2020); Li et al. (2021)

Table 3 (continued)

Perspectives		Critical factors	References
Context-related factors		Expertise and Knowledge	Alhazmi (2019)
		Third-party Assessment and Recognition	Triberti et al. (2020)
		Leadership and Management	Łapińska et al. (2021); Li et al. (2021); Ryan (2020)
		Lawful and Ethical Use	Bejger & Elster (2020); Chen & Wen (2021); Crigger et al. (2022); Hong & Williams (2019); Jain et al. (2020); Jelovac et al. (2021); Kraus et al. (2021); Qin et al. (2020); Ryan (2022); Sheridan (2019); Shin (2021); Sutrop (2019); Thiebes et al. (2021); Winfield & Jirotko (2018); Wirtz et al. (2019)
		Utilitarian Features	Ashfaq et al. (2020); Bilgihan (2016); Crigger et al. (2022); Gupta et al. (2021); Li et al. (2021); Pitardi & Marriott (2021); Moussawi et al. (2021); Nordheim et al. (2019); Saßmannshausen et al. (2021); Sheridan (2019); Triberti et al. (2020); Vassilikopoulou et al. (2018); Vourgidis et al. (2018); Yen & Chiang (2021)
		Hedonic Features	Ashfaq et al. (2020); Bilgihan (2016); Hasan et al. (2021); Kerassidou (2020); Langer et al. (2019); Li et al. (2021); Lou & Yuan (2019); Moussawi et al. (2021); Nordheim et al. (2019); Sheridan (2019); Triberti et al. (2020); Yen & Chiang (2021)
	Social Factors	Social Norms and Morals	Alaieri & Vellino (2016); Banks (2020); Crigger et al. (2022); Fan et al. (2020); Fang et al. (2022); Gillath et al. (2021); Janssen et al. (2020); Jelovac et al. (2021); Kuipers (2018); Lobschat et al. (2021); Sheridan (2019); Sullivan et al. (2020)
		Laws and Regulations	Jelovac et al. (2021); Lobschat et al. (2021)
		Culture and Values	Faisal et al. (2016); Hallikainen & Laukkanen (2018); Jelovac et al. (2021); Lei & Rau (2021); Lobschat et al. (2021); Robinson (2020); Schmidt et al. (2020)
	User-related Factors	Familiarity and Prior Experiences	Bawack et al. (2021); Chong et al. (2022); Gillath et al. (2021); Hafizoğlu and Sen (2018); Kaplan et al. (2021); Khaksari & Keyvanpour (2019); Qin et al. (2020); Saßmannshausen et al. (2021); Sheridan (2019); Sutrop (2019)
		Knowledge, Competency, Expectancy, and Expertise	Kaplan et al. (2021); Saßmannshausen et al. (2021); Starke et al. (2021)
		Demographic Background	Araujo et al. (2020); Chen & Wen (2021); Dos Santos et al. (2019); Gillath et al. (2021); Lou & Yuan (2019); Kaplan et al. (2021); Sheridan (2019)
		Big Five Personality Traits	Bawack et al. (2021); Gillath et al. (2021); Kaplan et al. (2021); Langer et al. (2019); Li et al. (2021); Oksanen et al. (2020); Sharan & Romano (2020)
		Propensity or Disposition to Trust	Agag & El-Masry (2017); Fan et al. (2020); Hallikainen & Laukkanen (2018); Kaplan et al. (2021); Kim & Peterson (2017); Langer et al. (2019); Nordheim et al. (2019); Qin et al. (2020); Saßmannshausen et al. (2021); Wei et al. (2019)
		Existing Perceptions	Araujo et al. (2020); Følstad et al. (2018); Gillath et al. (2021); Kaplan et al. (2021); Mou et al. (2017); Nordheim et al. (2019); Pappas (2016); Park (2020); Starke et al. (2021); Sullivan et al. (2020); Sutrop (2019); Wirtz et al. (2019); Yang & Zhang (2020); Yang et al. (2015)
Consequences of Trust in AI	Cognitive and Affective Changes	Change of Perception	Bedué & Fritzsche (2021); Vasiljeva et al. (2021)
		Change of Attitude and Feeling	Kim & Peterson (2017); Lee & Chen (2022); Vasiljeva et al. (2021)
		Subsequent Experiences and Satisfaction	Ameen et al. (2021); Hsiao & Chen (2021); Kim & Peterson (2017)
	Behavioral Changes	Intention to Adopt or Use	Bedué & Fritzsche (2021); Chatterjee & Sreenivasulu (2021); Fan et al. (2020); Kim et al. (2021); Meyer-Waarden & Cloarec (2022); Mostafa & Kasamani (2021); Moussawi et al. (2021); Park (2020); Priyadarshinee et al. (2017); Rahman et al. (2021); Starke et al. (2021); Vourgidis et al. (2018)
		User Commitment	Hsiao & Chen (2021); Lv et al. (2022); Jiang & Lau (2021)

assessed further from two specific perspectives: (i) trust in the technology and (ii) trust in the organization (Łapińska et al., 2021; Thiebes et al., 2021). The former includes users' trust in AI's various technological components, including the device, infrastructure, data transmission process, and analytic functions, while the latter is institution-based and focuses on the provider of AI-enabled services (Ahmadi & Tutschku, 2017; Gillath et al., 2021; Söllner et al., 2016; Wei et al., 2019). The ground for grouping the concept of trust into two individual categories is that people's trust in technology is different from their trust in a human, organization, or community (Schmidt et al., 2020). Compared to trust towards human beings, "users tend to have less trust towards AI by default where people are especially averse to algorithmic forecasters after seeing them perform, even when they see them outperform a human forecaster" (Schmidt et al., 2020, p.4). This approach allows for more precise measurements of the components of trust. For instance, the ability competent of trust in AI technology is based on the characteristics of the AI-enabled service and its provider (Mal et al., 2018; Nundy et al., 2019). Thus, these characteristics should include not only the AI technology's reliability and quality (e.g. artifact design and safeness) but also the organization's (i.e. service provider) capability, management, strategy, and culture (Li et al., 2021; Mal et al., 2018). Meanwhile, the benevolence component of users' trust in the AI technology is reflected by its genuine communications with users and care for their needs. It also requires the organization's care towards its stakeholders, such as its employees, partners, the environment, and society. Furthermore, the integrity component of users' trust depends on whether the AI technology is able to keep promises and provide fair and transparent services (Crigger et al., 2022; Crockett et al., 2020; Felzmann et al., 2019; Qin et al., 2020; Ryan, 2022; Shin, 2021). It also requires both the AI-enabled service and its provider to follow the norms and regulations (Gillath et al., 2021; Janssen et al., 2020; Kuipers, 2018). Thus, some researchers use the term "digital trust" that highlight the organizational and technical aspects to suggest the importance of CDR for user trust building (Jelovac et al., 2021). Digital trust suggests users' trust should be not only adopted as a universal concept but related to specific fields (e.g. digital technologies).

Further, Li et al. (2021) also related users' trust in an organization to its stakeholders, namely through the aspects of trust in other related organizations and trust in people in the organization. The former concerns users' attitudes towards other relevant organizations within the supply chain of the AI service provider, for example, based on the information technology integration with its supplier(s). The latter refers to the user's perception of the behaviors of employees and leaders in the related organizations. In this context, scholars have highlighted the importance of an

organization's AI promoters, who are responsible for introducing, implementing, and promoting the new technology.

Thus, users' trust in AI depends on their general trusting beliefs, trust in the associated organization, and trust in the technology. Thiebes et al.'s (2021) study illustrated the relationships between three factors: (i) a user's general trusting beliefs in other people and new technologies will affect their (ii) trust in an organization's adopted AIs, which will finally determine their (iii) trust in specific people (such as AI promoters) and the AI applications adopted by the organization. In this regard, Söllner et al.'s (2016) study highlighted the unique importance of trust in the organization as it significantly affects users' trust in the technology. Further, Mou et al. (2017) argued that trust in the organization, compared to trust in the technology, plays an even more significant role in users' acceptance of services supported by new technologies.

Cognitive, affective, and behavioral trust

The above discussions suggest those studies on users' trust tends to focus on the trusted objects (e.g. technologies and organizations). However, the formation of trust should also depend on the components related to users. For instance, trust in technology and the organization originates from the qualitative characteristics (e.g. competence, ability, and integrity) perceived by the users (Khaksari & Keyvanpour, 2019; Kraus et al., 2021; Shi et al., 2021). Thus, some researchers have classified this as the cognitive aspect of trust as it is generally based on the user's rational cognition and calculation of losses and gains of trusting the technology (Gillath et al., 2021; Glikson & Woolley, 2020; Khaksari & Keyvanpour, 2019; Kraus et al., 2021). The formation of cognitive trust is often related to the reputation of the service provider organization in terms of the characteristics believed or communicated by other users. This aspect of trust is also role-based, as it originates from a consideration of the AI's role in service (Khaksari & Keyvanpour, 2019).

In addition to the cognitive aspect, scholars believe that user trust should also be studied from affective and behavioral perspectives. The affective (also known as emotive) aspect of trust focuses on users' intellectual background and their feelings and attitudes towards AI. Compared to cognitive trust, the affective aspect is irrational due to its dispositional nature, which means that trust formation is significantly related to users' psychological backgrounds, even if they are unfamiliar with AI (Gillath et al., 2021; Glikson & Woolley, 2020; Khaksari & Keyvanpour, 2019; Kraus et al., 2021; Shi et al., 2021). In this situation, users' affective trust in AI is caused by their love or propensity to trust others, without identifying the technology or the service provider (Khaksari & Keyvanpour, 2019). Therefore, affective trust can lead to users making decisions based only

on their psychology while ignoring all the facts about AI and the related service providers (Khaksari & Keyvanpour, 2019). Last, behavioral trust is based on repeated interactions between the user and AI applications over a period of time. Compared to affective trust, this type is based on a user's direct and repeating experience with the identified service provider and technologies (Khaksari & Keyvanpour, 2019). Thus, these two types of trust differ in terms of the basis of identification and relationship (Khaksari & Keyvanpour, 2019; Li et al., 2021). Furthermore, behavioral trust is a form of direct trust, whereas cognitive trust may arise from other people's recommendations. Therefore, the difference between behavioral and cognitive trust is the acquisition method involved (Li et al., 2021).

Trust in specific AI applications and contexts

The above discussions suggest that users' trust in AI should be studied as a multiple-component concept. First, people may use similar criteria when forming trust in different objects. Therefore, users' trust in AI should include the universal components (e.g. predictability, dependability and faith) that can also be applied to other trusted objects. Thus, these components reflect the generic requirements for trust building. Second, AI has been adopted by different organizations as new digital technology. Therefore, users' trust in AI should also include the components of trust related to the technology and the organization. These components provide user trust, as a generic concept, with a focus on AI-supported service providers. Third, from the users' perspective, they may form trust in AI based on their different knowledge, psychology, and interactions related to the technology and the organization. This accordingly suggests that the components of trust in AI should also include users' cognitive, affective, and behavioral aspects. Further, it is noticeable that some scholars have limited the scope of users' trust to specific AI applications. For instance, Park (2020) used the term "Trust in service robots" in his study on tourism management. Meanwhile, Qin et al. (2020) adopted the term "Trust in AI-based educational systems" for their research on elementary and secondary education. The adoption of AI technology in a wide range of sectors and contexts implies that a person's trust in different AI applications may vary. This can be explained by the fact that users may have different levels of involvement with AI applications in various contexts (Mostafa & Kasamani, 2021). For example, even though a user accepts chatbots for customer services, he/she may not have the same level of trust toward self-driving cars, because the latter applies AI to a more concerning context. However, the above-mentioned three classes of user trust components have not considered the users' inconsistent trust in various contexts of AI applications. For different applications, users may have to use different evaluation criteria

for AI's performance, process, and purpose, have different expectations for the technology and the organization, and need different levels of knowledge and interactions. Based on this, we proposed an additional component of users' trust in AI: the trust in specific applications. This component suggests that users may not necessarily expand their trust in one AI application to others. Therefore, even when a trusted organization applies the same technology to a new AI-supported product or service for the same users, it still needs to adjust the trust-building tactics.

Critical factors influencing trust in AI

The existing studies based on the UTAUT model suggest that users' trust in AI is dependent on performance expectancy, effort expectancy, and social influence (Fan et al., 2020). Furthermore, trust is a synthesis of multiple components that can be studied from different perspectives. This implies that the formation of trust is affected by various factors. From the organizational environment perspective, influencing factors include the internal (e.g. technologies and strategies adopted by an organization) and external (e.g. industry context, social and cultural influences, and customers/users) aspects (Everson, 2003). For example, Qin et al. (2020) and Kaplan et al. (2021) classified antecedents of users' trust as technology-related, context-related, and user-related. Other studies also related users' trust to organizational (Li et al., 2021) and social aspects (Kuipers, 2018). Meanwhile, the CDR framework suggests that the influences include organization factors, customer factors, and social factors (Elliott et al., 2021; Lobschat et al., 2021; Weber-Lewerenz, 2021). Thus, the present study divides the influencing factors of users' trust in AI into five major groups, i.e. technological, organizational, contextual, social, and personal factors.

Technology-related factors

Trust in technology is an essential component of user trust and the basis of people's perceptions of AI technologies and applications. For instance, studies based on the UTAUT model suggest that people's trust in AI is driven by its perceived quality in terms of compatibility, ease of use, and performance (Mostafa & Kasamani, 2021). Thus, technology characteristics affect effort expectancy and influence users' trust in AI (Fan et al., 2020). However, the quality of a compatible, user-friendly and high-performing AI application is reflected by multiple facets. Thus, some scholars, such as Nasirian et al. (2017), explored the antecedent of trust by considering users' perceptions of the service quality, information quality, and system quality of AI applications. A few studies have also considered perceived designed quality as a technology-related factor (Kim & Peterson, 2017; Nilashi et al., 2015).

First, AI technologies are often applied to service sectors (Balakrishnan & Dwivedi, 2021). Therefore, users' perception of service quality is an important factor in building trust (Ameen et al., 2021; Ashfaq et al., 2020; Bedué & Fritzsche, 2021; Hsiao & Chen, 2021; Kim & Peterson, 2017; Nasirian et al., 2017; Oliveira et al., 2017; Wibowo et al., 2020). Users evaluate service quality by comparing their perception of the service received with their expectation of what that service should offer (Ameen et al., 2021; Sheridan, 2019). A service provider's effort for providing high-quality services is often seen by users as a sign of its capability to display integrity and trustworthiness (Kim & Peterson, 2017; Wibowo et al., 2020). Researchers consider service quality a technology-related factor because it requires the support of the more advanced AI heuristics and algorithms, such as automatic improvements through machine learning based on experience and data use (Sheridan, 2019). In the context of AI-enabled services, users often require convenient and safe performances and have additional expectations, such as more speedy responses, a better understanding of user preferences, and additional customer support, compared to traditional services (Ameen et al., 2021; Sheridan, 2019). Moreover, AI technologies should not just focus on individuals' pre-purchase activities if they are to provide high levels of service quality, but they also need to support the customers after their transactions (Nasirian et al., 2017; Nilashi et al., 2015). Balakrishnan and Dwivedi (2021) used the concept of cognitive absorption to reflect on the quality of AI-enabled services. Specifically, a high-quality service allows users to control their interactions, piques their curiosity, offers them enjoyment, immerses them in the tasks, and even makes them lose track of time. Further, to provide high service quality, AIs must also demonstrate their empathy and compassion for users, which improves emotional trust (Kerasidou, 2020; Lv et al., 2022). Some researchers, such as Ameen et al. (2021), also related the AI-enabled service quality to users' perceived quality of the information system and its provided content.

Second, perceived information quality depends on the contents provided by an AI application to its users (Ashfaq et al., 2020; Bedué & Fritzsche, 2021; Nasirian et al., 2017). Thus, a user's trust in the AI application depends on whether it is able to provide accurate, factual, current, and complete content that supports the user's tasks (Ahmadi & Tutschku, 2017; Kim et al., 2021; Kim & Peterson, 2017; Nasirian et al., 2017; Nilashi et al., 2015; Yen & Chiang, 2021). This high-quality content should also be based on traceable and analytical data (Ahmadi & Tutschku, 2017). When users communicate with an AI, they usually expect the conversations to be informative and avoid too many irrelevant messages (Yen & Chiang, 2021). Moreover, AI-enabled information systems should avoid inappropriate content, such as offensive language (Hong & Williams,

2019). It is also essential to avoid providing biased or discriminating information (Thiebes et al., 2021). For example, according to Hong and Williams (2019), some AI-enabled crime prevention applications suggest that black prisoners have a higher likelihood of future crimes than other prisoner groups with similar criminal records, despite the lack of evidence. This prejudice caused by the biased automation of AI can be considered a form of the "automating inequality" issues that have drawn the attention of researchers in the recent years (Eubanks, 2018; Mañero, 2020). These issues suggest that the increasing automation of AI can worsen the inequality for disadvantaged groups (Eubanks, 2018). Some researchers attribute this to the human bias because AI applications are fundamentally built on data generated by humans (i.e. data scientists, engineers and users) or collected via systems and algorithms designed by humans (Ntoutsis et al., 2020; Sanclemente, 2021). Meanwhile, Lobschat et al. (2021) argue that the biased projection may be unintendedly caused by using machine learning algorithms based on historical data alone. To reduce the bias, it is suggested to enhance the diversity of data sets and the inclusion of multiple groups in algorithm development (Ntoutsis et al., 2020; Sanclemente, 2021). Further, AI-powered organizations should be aware of the longer-term mutability of their digital assets during the decision-making and operational processes (Lobschat et al., 2021).

Third, the system quality of an information technology application can be evaluated in terms of the reliability, flexibility, accessibility, and timeliness of its functional features (Bedué & Fritzsche, 2021; Lazanyi, 2018; Nassar et al., 2020). Thus, when an information system is easy to access and contains no errors (e.g. miscalculations, inaccuracies, misinterpretations, overestimations or underestimations), it indicates its trustworthiness (Kim & Peterson, 2017; Triberti et al., 2020). The perceived system quality often concerns security issues and whether the AI-supported system has an established structural assurance, e.g. site authentication and privacy protection policies and regulations (Anwar, 2020; Chatterjee & Sreenivasulu, 2021; Kim & Peterson, 2017; Nasirian et al., 2017; Nilashi et al., 2015; Qin et al., 2020). In other words, users will have more confidence in an information system's trustworthiness if it has the mechanisms to protect their shared personal information (e.g. identity, location and device), and provides the assurance that their sensitive data will be accessed and viewed only by authorized people (Ahmadi & Tutschku, 2017; Kim & Peterson, 2017; Robinson, 2020). Thus, a high-quality system should aim to confidentially, integrally, and authentically transmit data, which still requires more effort in on-going improving the transparency, tractability, and predictability of its processes (Felzmann et al., 2019; Girsang et al., 2020; Hepworth et al., 2020; Nassar et al., 2020; Qin et al., 2020; Shin, 2021). For example, AI algorithm developers should also recognize

their ethical responsibility at the post-implementation stage, which requires continuous engagement and monitoring in the long term (Lobschat et al., 2021).

Fourth, a number of studies highlight the importance of the perceived design quality of AI-supported information systems. This type of quality perception can be enhanced by using aesthetic features of interfaces (e.g. color schemes) that are pleasing and attractive to users (Ameen et al., 2021; Faisal et al., 2016; Hsiao & Chen, 2021; Nilashi et al., 2015; Qin et al., 2020). A visually appealing design reflects the service provider's capability and professionalism, which can contribute to the formation of user trust (Følstad et al., 2018; Kim & Peterson, 2017). An AI-supported information system should be understandable and personalized, allow users to navigate the content easily, and have the capability to support multimedia in order for its design to be considered of high quality (Anwar, 2020; Moussawi et al., 2021; Nilashi et al., 2015; Qin et al., 2020; Shi et al., 2021). Some scholars, such as Faisal et al. (2016), emphasize the importance of a user-friendly design. For example, the content displayed in an AI-supported interface should be easy to read, especially for visually impaired users.

Notably, the importance of anthropomorphism or humanness has gained attention in a significant number of recent studies (Bernotat et al., 2017; Følstad et al., 2018; Gulati et al., 2019; Hsiao & Chen, 2021; Kaplan et al., 2021; Kim & Song, 2021; Lazanyi, 2018; Lee & Chen, 2022; Li et al., 2021; Lu et al., 2022; Mohanty, 2020; Moussawi et al., 2021; Pelau et al., 2021; Shi et al., 2021; Troshani et al., 2020; Yen & Chiang, 2021). This is because the anthropomorphism of AI is believed to foster users' emotional trust (Gulati et al., 2019; Shi et al., 2021). Troshani et al. (2020) defined anthropomorphic characteristics in terms of the ability to have a presence, make human connections, show emotion, emulate human communication, and take the initiative to help. Researchers also relate it to human features such as body shapes, facial expressions and small talk (Bernotat et al., 2017; Paradedda et al., 2016). This concept is related to the social presence theory, which refers to the subjective experience of being present with a natural person and having access to their thoughts and emotions. Thus, the "sense of being with another" (i.e. human warmth, sensitivity, and sociability) depends on the design of interfaces in human-AI interactions (Yen & Chiang, 2021; Zhang et al., 2021). Further, an AI's anthropomorphism level is supported by its intelligence, including its ability to innovate, understand the subtleties of human interactions, solve problems, and consistently achieve expected outcomes (Pelau et al., 2021; Troshani et al., 2020).

Organizational factors

As previously discussed, individuals' trust in organizations (i.e. institution-based trust) is another essential component of the concept of user trust that complements their trust in

technologies (Chen & Wen, 2021; Wei et al., 2019). Therefore, it is important to study organization-related factors as antecedents of trust, and a number of studies have focused on the image and reputation of AI-enabled service providers (Burda & Teuteberg, 2014; Hamid et al., 2015; Kaplan et al., 2021; Khaksari & Keyvanpour, 2019; Oliveira et al., 2017; Qin et al., 2020; Xin et al., 2015). Organizational factors are related to social evaluation, which is defined as the degree of an organization's compliance with societal norms and regulations (Hamid et al., 2015; Jelovac et al., 2021; Lobschat et al., 2021; Ryan, 2020). Thus, an organization's reputation can be represented by its social and digital responsibility, e.g. having environmentally friendly policies, supporting local communities, treating employees fairly, and ethical data use (Elliott et al., 2021; Jelovac et al., 2021; Liyanaarachchi et al., 2020; Lobschat et al., 2021; Nguyen & Pervan, 2020; Weber-Lewerenz, 2021). The effect of organizational reputation on users' trust is especially significant in the case of e-commerce (Oliveira et al., 2017). This may be because an organization's reputation reflects its time and effort to offer sincere customer care. E-commerce service providers are usually recognized by their brand identities (organization names, trademarks, and symbols). Users in the e-commerce context may consider brand identities as substitutes for the information of a purchased service during the decision-making process. Therefore, a reputable brand tends to gain more consumer trust (Söllner et al., 2016).

The reputation of a service provider is often shared among users and, therefore, plays a significant role in fostering trust; this also extends to AI (Nordheim et al., 2019). To build user trust, organizations should demonstrate integrity, especially in data protection against misuse, which has become a public concern for AI technologies (Pickering, 2021; Weitz et al., 2019). Therefore, organizations need to develop and maintain relevant policies, principles, and procedures, including ethical committees, risk assessments, and regular audits (Janssen et al., 2020). A trustworthy service provider also needs to have a healthy organizational culture, ensuring the adherence to professional norms, perception of data as an asset, and implementation of rewards and sanctions (Jain et al., 2020; Wibowo & Mubarak, 2020). A user's trust in a service provider is also based on the latter's expertise and knowledge of its users, and timely and regular communication with customers (Alhazmi, 2019). Furthermore, reputation is often based on third-party assessment and recognition. In other words, people's decisions are usually dependent on others' opinions regarding the trustworthiness of the object. Therefore, gaining testimonials from a recognized expert or authority in the field can help an AI-enabled organization to be trusted and accepted (Triberti et al., 2020).

Moreover, AI technologies can not only support an organization's customer service (e.g. chatbots) but may be

applied to other management functions, such as production and manufacturing. Thus, the employees of the organization would also be users of AI technologies in this scenario (Łapińska et al., 2021; Li et al., 2021; Ryan, 2020). Accordingly, some researchers have focused on the trustworthiness of AI technologies within organizations. For instance, the commitment of an organization's leaders (i.e. management commitment) to the implantation of AI in the organization is a critical factor. Leaders with a strong will to adopt AI in the organization are more likely to have solid strategic visions, invest greater resources, and set more incentivizing rules (Li et al., 2021). Furthermore, employees' trust in AI is also related to the organizational culture. For instance, Li et al. (2021) argued that authoritarian culture can support trust in AI within the organizations because the autocratic leadership involved may motivate subordinates to trust in AI applications to maintain the consistency of ideas with their leaders.

Considering the above, the organizational approach explores the role of AI-enabled service providers' characteristics in user trust (Chen & Wen, 2021). From the CDR perspective, studies in this field have stressed the significance of the lawful and ethical use of AI technologies (Bejger & Elster, 2020; Lobschat et al., 2021). First, the adoption of AI requires the assurance of fairness and honesty and the avoidance of discrimination (e.g. sexism or racism) against any user group (Bejger & Elster, 2020; Crigger et al., 2022; Hong & Williams, 2019; Jain et al., 2020; Qin et al., 2020; Sheridan, 2019; Shin, 2021; Thiebes et al., 2021; Wirtz et al., 2019). Therefore, AI-enabled organizations must consider the diversity of user groups and their abilities, skills, and needs to remain ethical and trustable (Bejger & Elster, 2020; Hong & Williams, 2019). Second, the use of AI technologies should respect human autonomy (Crigger et al., 2022; Jain et al., 2020; Kraus et al., 2021; Qin et al., 2020; Ryan, 2020). Consumers may be concerned about the virtual relationship between AI applications and users, and expect humans to be in the decision loop. Thus, organizations should allow users to make their own choices and have complete control over their data (Bejger & Elster, 2020; Sheridan, 2019). Third, AI-enabled organizations should use the relevant technologies to further their social responsibility and support environmental sustainability (Bedué & Fritzsche, 2021; Bejger & Elster, 2020; Lobschat et al., 2021; Ryan, 2020). Therefore, organizations should consider the beneficence of using AI technologies, which requires balanced costs and benefits for their stakeholders. For example, organizations should allow users to recognize how their data are collected, used and stored, and provide them with valuable returns, e.g. monetary compensation and improved services (Lobschat et al., 2021). Ensuring the lawful and ethical use of AI technologies in line with the above does not only require secure, reliable, and robust AI algorithms but accountable culture and transparent governance

and culture of the organization (Bejger & Elster, 2020; Crigger et al., 2022; Felzmann et al., 2019; Lobschat et al., 2021; Sheridan, 2019; Sutrop, 2019; Winfield & Jirotko, 2018). For example, to protect users' data, it expects AI-enabled organizations to adopt encryption technologies, even though more investments are needed. It also requires the established CDR culture, in which AI application developers are responsible for consequences that arise from their duties (Lobschat et al., 2021).

Context-related factors

The role of organizational factors may vary because organizations apply AI technologies in different contexts (e.g. industry context) (Araujo et al., 2020; Kolasinska et al., 2019; Lobschat et al., 2021; Wirtz et al., 2019). This can also be attributed to the fact that people may use different AI-enabled applications for various purposes and tasks (Ahn et al., 2022). The task complexity affects performance expectancy, influencing users' trust in AI (Fan et al., 2020). Thus, it is also necessary to consider the effects of various features of AI-enabled services/products on users' trust in different contexts. Many studies in this field apply the UTAUT/UTAUT2 to different contexts. This theory suggests that people's acceptance of new technology will depend on their perception of its utilitarian features (e.g. perceived usefulness and ease of use) and hedonic motivation (perceived enjoyment) (Ashfaq et al., 2020; Moussawi et al., 2021). Bilgihan (2016), too, argued that the utilitarian and hedonic elements of a product (or service) will affect users' trust in it.

The utilitarian aspect concerns how an AI-supported application can offer value to users' task-related and rational consumption (Bilgihan, 2016). Existing studies based on UTAUT models highlight certain utilitarian features (i.e. perceived usefulness and ease of use) as the driving factors of user trust in different contexts (Zhang et al., 2021). However, users' perceptions of the utilitarian value can differ across the various contexts of AI usage. For instance, explainability, which is the ability to interpret and justify the causal relationships between different factors, is essential for news recommendation systems or medical computer systems (Crigger et al., 2022; Li et al., 2021). User trust in chatbots will depend on the AI applications' expertise, which is reflected by their knowledge of service context and user needs (Nordheim et al., 2019; Yen & Chiang, 2021). The utilitarian values of AI are also based on the error costs or the perceived severity in different contexts (Saßmannshausen et al., 2021; Sheridan, 2019; Vassilikopoulou et al., 2018). A higher level of perceived severity will make it more challenging to develop user trust (Vassilikopoulou et al., 2018). For example, it is typically considered a severe situation when people apply AI technologies to vehicles or medical expert systems because of the direct effects on their

safety and health (Gupta et al., 2021; Pitardi & Marriott, 2021; Vourgdis et al., 2018).

The hedonic aspect is based on users' subjective experiences of fun and playfulness (Bilgihan, 2016). For example, consumers may perceive the hedonic value of AI in terms of how it entertains them while providing a service (Lou & Yuan, 2019). The hedonic experience can also be related to the novelty of AI technologies. For instance, a study by Hasan et al. (2021) on voice-controlled AI applications revealed that users perceive a novelty value when the technology provides unique experiences that satisfy their curiosity. Users may demand different hedonic values of AI technologies in different contexts. For example, Li et al.'s (2021) study revealed that the consistency and personality of voice are important for in-vehicle assistants and smart speakers. Further, anthropomorphic characters (e.g. with the capability to interact in a natural language), social interactions (Shin, 2021), and playfulness are believed to be important factors for building users' trust in chatbots (Nordheim et al., 2019; Yen & Chiang, 2021). Emotional factors (e.g. anxiety) affect users' trust in self-driving vehicles and medical diagnostic aids (Li et al., 2021). Human-like representation in terms of appropriate emotional expressions and matching users' personalities may also be required for AI applications in service contexts, such as social robots (Kerasidou, 2020; Langer et al., 2019). In the case of AI-supported appliances, users may consider aesthetic appearances a hedonic value (Sheridan, 2019).

In practice, integrating both utilitarian and hedonic aspects can be more efficient in building user trust. Triberti et al. (2020) highlighted exploitability as an important utilitarian feature, i.e. interpreting the processes and outcomes in intelligible ways for users who are supposed to benefit from the AI's support. They also emphasized the necessity to develop the hedonic features of AI devices and interfaces to ensure that they are entertaining, engaging, mildly challenging and allow personalization.

Social factors

People, organizations and industries are components of the social environment. Thus, social factors such as social norms and culture are traditionally considered in the domain of human interactions. With the trend of AI becoming more intelligent, people's trust in AI will increasingly be akin to that of trusting another human. Hence, the roles of social norms and culture will also be more significant in trusting an AI (Crigger et al., 2022; Fan et al., 2020; Fang et al., 2022; Kuipers, 2018; Sheridan, 2019). Some AI applications (e.g. chatbots) have been seen as social actors with the capacity to possess human-like attitudes or feelings. Therefore, users are likely to apply some social rules, such as manners, when interacting with AIs. Furthermore, the concept of user trust

includes the component of integrity, which requires both an AI-enabled service and its provider to follow societal norms. Thus, several studies have explored the social factors related to building user trust in AI (Gillath et al., 2021; Janssen et al., 2020; Kuipers, 2018; Sheridan, 2019). For example, trust in an AI application (e.g. robots) would depend on its compliance with social norms and morals, e.g. care, fairness, loyalty, and purity/sanctity (Alaieri & Vellino, 2016; Banks, 2020; Sullivan et al., 2020). Notably, a higher requirement for purity may reduce the level of religious groups' trust in AI applications because the technologies may be considered intrinsically impure due to being artificial constructs that lack human souls and hearts (Banks, 2020). The effect of social rules may depend on whether they have become clear laws and regulations. For example, people are more likely to trust AI when users' consent is a well-enforced legal requirement for data collection (Jelovac et al., 2021; Lobschat et al., 2021).

As another important social factor, culture can directly influence people's perceptions, behaviors, and preferences in any nation (Robinson, 2020). Therefore, a stream of research focuses on relating social factors to cultural variance in different countries or markets. They aim to determine how the features of national cultures (i.e. individualism/collectivism, uncertainty avoidance, long-term/short-term orientation, power distance, and masculinity/femininity) affect people's disposition to trust AI (Faisal et al., 2016; Hallikainen & Laukkanen, 2018; Robinson, 2020). First, as one of the cultural dimensions, individualism and collectivism describe the relationships of individuals within a nation or cultural group. In individualist cultural groups, people tend to focus on themselves and their immediate family, while collectivism values the strong ties between individuals in society. Thus, people from collectivist groups are more likely to trust others, especially their ability (Hallikainen & Laukkanen, 2018; Robinson, 2020). Second, uncertainty avoidance is the extent to which individuals perceive and react to threats due to unknown situations and ambiguity (Hallikainen & Laukkanen, 2018; Robinson, 2020). People often experience uncertainties when adopting a new technology because the outcome is unknown. Therefore, uncertainty avoidance reduces users' disposition to trust (Hallikainen & Laukkanen, 2018; Robinson, 2020; Schmidt et al., 2020). People from a country with a higher level of uncertainty avoidance will attach more importance to the security and simplicity of new technology and its applications (Faisal et al., 2016). Third, long-term oriented groups are likely to focus on the future, whereas short-term oriented groups emphasize the importance of the present and past (Hallikainen & Laukkanen, 2018; Robinson, 2020). Thus, long-term oriented people tend to value the long-term trust that is built on not only the ability but also the consistency of benevolence and integrity of the trustee (Hallikainen & Laukkanen, 2018).

Fourth, power distance refers to people's values regarding status, authority, and power in organizations and society. A culture group with high power distance tends to expect the leaders to have the authority and ability to make the right decisions regarding the adoption and implementation of AI technologies (Lei & Rau, 2021). Power distance also increases the ethical expectations placed on leaders' decisions. Accordingly, it affects the benevolence and integrity components of user trust. Last, a masculine culture is action-oriented and focuses on achieving goals, while a feminine culture attaches more importance to harmonious relationships among the people. It, therefore, implies that feminine cultural groups tend to believe that "most people can be trusted" (Hallikainen & Laukkanen, 2018).

User-related factors

As discussed in the previous section, studies on social factors (e.g. national cultures) explore how users' trust in AI is affected by society at the macro level. However, users' trust is also affected by micro-level factors, i.e. their personal backgrounds and characteristics (Saßmannshausen et al., 2021; Starke et al., 2021). Scholars often relate user trust to the timing of the occurrence of trust (Li et al., 2021). For instance, Qin et al. (2020) and Sheridan (2019) argued that users' familiarity and prior experiences with the technology can affect their tendency to trust. It implies that positive past experiences reduce the user's difficulty in trusting an AI technology, while negative prior experiences (such as privacy issues) make the process more difficult (Lobschat et al., 2021). This viewpoint is consistent with the findings of studies by Bawack et al. (2021), Gillath et al. (2021), Khaksari and Keyvanpour (2019) and Söllner et al. (2016). Further, Sutrop (2019) stated that prior experiences are a type of empirical knowledge; it means people gain knowledge from their experience with an object. In the same vein, Kaplan et al. (2021) and Saßmannshausen et al. (2021) classified past experience as an element of ability-based antecedents of trust, which is also related to users' knowledge, competency, expectancy, and expertise (i.e. expert status). Trust is required when a user has imperfect knowledge concerning a new technology (Starke et al., 2021). Thus, trust is unnecessary for users with complete knowledge about new technology, whereas those without knowledge, competency, or expertise cannot reasonably and thoroughly build trust. Those without any direct prior experience of using AI may rely on their recent experience with other new technologies, especially those offered by the same provider (Langer et al., 2019). Therefore, the users' trust in AI can be built on direct or indirect experiences. Chong et al. (2022) and Hafizoğlu and Sen (2018) related the role of prior experiences in trust forming to its effect on users' emotions, attitudes,

confidence, and expectations. For instance, previous negative experiences may lead to users being more stressed or cautious, making it more difficult for them to rely on AIs. A mismatch between a user's expectation and the reality of AI applications can negatively affect trust (Wirtz et al., 2019). However, Hafizoğlu and Sen's (2018) study suggests that prior experience with an AI-enabled service provider can only affect the initial trust. In other words, the effect of prior experience on trust diminishes over time.

User-related factors also include the characteristics of users' demographic backgrounds and personality traits. The education level, socioeconomic status, age/generation, and other demographic characteristics likely affect users' understanding, adoption, and use of new technologies (Araujo et al., 2020; Lou & Yuan, 2019; Sheridan, 2019). For instance, Kaplan et al. (2021) reported that several demographic factors (i.e. age, education, and gender) can affect users' trust in AI, of which gender is the most influential one. The authors suggested that male users are more likely to trust AI than female users. This finding is consistent with a study by Dos Santos et al. (2019) on medical students' attitudes toward AI. Some researchers argued that females are more likely to be anxious and risk-averse before accepting new technologies; therefore, females tend to have more security and privacy concerns when adopting new applications (Shao et al., 2020). As for the effect of age, it has been found that younger people are more likely to trust AI than older groups (Gillath et al., 2021). This is often attributed to the lower level of digital literacy and experience, and the higher likelihood of visual, physical and cognitive impairments among the older groups. It increases the difficulty of processing the information regarding new technologies and operating the new applications, which can accordingly reduce the trust level (Hua et al., 2021; Liyanaarachchi et al., 2020). Additionally, people's educational level is positively associated with their trust in AI (Chen & Wen, 2021). This can be explained by the fact that higher education provides people with more knowledge of risk management (Jung, 2015), which may reduce their aversion to the risks of adopting new technologies. From the CDR perspective, AI-powered organizations should adopt a more user-centered approach to provide customized solutions (e.g. additional training and support services) to address vulnerable groups' concerns and needs (Liyanaarachchi et al., 2020; Weber-Lewerenz, 2021).

Furthermore, a number of studies have underlined the significant role of users' personalities in their trust in AI (Bawack et al., 2021; Gillath et al., 2021; Langer et al., 2019; Li et al., 2021; Oksanen et al., 2020; Sharan & Romano, 2020). These studies commonly adopted the Big Five model to assess the personality traits of AI users; this model measures the dimensions of openness, conscientiousness, extraversion, agreeableness, and neuroticism (Bawack

et al., 2021; Gillath et al., 2021; Langer et al., 2019; Li et al., 2021; Oksanen et al., 2020; Sharan & Romano, 2020). Openness is the term for “openness to experience”, which can be defined as an individual’s propensity to be intellectually curious and open to ideas (Bawack et al., 2021; Li et al., 2021; Oksanen et al., 2020; Sharan & Romano, 2020). In other words, those with high openness value novelty, appreciate art, tend to adopt imaginative and creative thinking styles, and enjoy seeking deeper meaning in things. Conversely, a low level of openness indicates a preference for familiarity and traditions (Bawack et al., 2021; Gillath et al., 2021; Langer et al., 2019; Li et al., 2021; Oksanen et al., 2020; Sharan & Romano, 2020). Conscientiousness is represented by people’s ability to control, regulate, and direct their impulses in their decisions and behaviors. A higher level of this personality trait suggests a preference for being cautious and organized, whereas a lower level indicates a tendency toward being flexible and casual (Bawack et al., 2021; Gillath et al., 2021; Langer et al., 2019; Li et al., 2021). Extraversion (also spelled as extroversion) describes people who tend to be talkative and fast-paced. They prefer to draw the attention of others. Conversely, introverted individuals are unlikely to start conversations and feel uncomfortable around people (Oksanen et al., 2020; Sharan & Romano, 2020). Agreeableness can be reflected by the importance of cooperation and social harmony. In other words, agreeable people tend to be more obliged, kind, sympathetic, and accommodating. Thus, they value maintaining positive relationships with others. In contrast, individuals with lower levels of agreeableness are less likely to compromise their interests for others and tend to challenge others’ opinions (Gillath et al., 2021; Langer et al., 2019; Oksanen et al., 2020). Neuroticism describes an individual’s emotional stability. People with a higher level of this personality trait tend to get stressed out and panic easily. By contrast, a lower level indicates the ability to control negative emotions such as anxiety, anger, and depression (Bawack et al., 2021; Langer et al., 2019; Oksanen et al., 2020; Sharan & Romano, 2020). Researchers in this field tend to agree that all the five personality traits are predictors of trust in AI. Specifically, people are more likely to trust AI when they have high levels of openness, conscientiousness, agreeableness, extraversion and a low level of neuroticism (Bawack et al., 2021; Kaplan et al., 2021). However, the effect of an extroverted personality on user trust in the AI is inconsistent in the existing research, with some studies finding that extraversion contributes to a higher level of trust while others show the opposite effect (Kaplan et al., 2021).

Moreover, the propensity or disposition to trust is often studied as a type of personality that affects users’ trust in technologies (Agag & El-Masry, 2017; Hallikainen & Laukkanen, 2018; Kaplan et al., 2021; Langer et al., 2019; Nordheim et al., 2019; Qin et al., 2020). This concept is defined as

the general tendency for an individual to trust others, which has a global effect on trust intentions (Agag & El-Masry, 2017; Fan et al., 2020; Hallikainen & Laukkanen, 2018; Kaplan et al., 2021; Langer et al., 2019; Qin et al., 2020; Wei et al., 2019). Stated differently, a higher level of propensity or disposition to trust indicates the greater tendency of an individual to trust other people. This tendency can also be applied to people’s trust in products and services. For example, Nordheim et al. (2019) used the term “propensity to trust technology” to study AI applications for customer services. People’s propensity to trust is affected by their personalities (Kim & Peterson, 2017; Langer et al., 2019) as well as their digital affinity, represented by the “degree to which the human has attraction, interest and pleasure in learning/using digital methods/technologies (Saßmannshausen et al., 2021, p. 1337). Similarly, Fan et al. (2020) adopted the concept of “personal innovativeness” to represent a person’s willingness to try out new technology. Personal innovativeness has a positive effect on effort expectancy and, accordingly, contributes to users’ initial trust (Fan et al., 2020).

Users’ trust in new technology is also driven by their existing perceptions. Risk is a necessary precondition that influences people’s trust in new technologies, including AI; therefore, the significant role of risk perception has been widely discussed in this field (Araujo et al., 2020; Crockett et al., 2020; Følstad et al., 2018; Gillath et al., 2021; Kaplan et al., 2021; Park, 2020; Starke et al., 2021; Sullivan et al., 2020; Sutrop, 2019; Yang et al., 2015). For example, people who perceive risks to their safety, privacy, or employment are less likely to trust AI (Wirtz et al., 2019). People may perceive various types of risks, including economic/financial, function/performance, physical/security, time/opportunity, privacy, psychological, and social (Gupta et al., 2021; Jiang & Lau, 2021; Sheridan, 2019; Triberti et al., 2020). First, economic risk refers to the possibility of financial loss due to using AI in relevant activities, such as investments or online transactions. Second, function risk is the potential that an AI-enabled service will not provide as valuable a performance as expected. Third, the physical risk is related to factors that can harm the user’s health or personal security. Fourth, time/opportunity risk is perceived when a user believes that their time and energy may be wasted by using an AI-enabled service or that they would miss out on the opportunity to use a better alternative. Fifth, privacy risk is the possibility of losing control over personal information. Sixth, social risk refers to the possibility of a user losing status in their social groups due to their failure to adopt AI-enabled services. Last, the psychological risk is the possibility of the above-mentioned risks affecting the user’s mood or causing anxiety (Gupta et al., 2021; Jiang & Lau, 2021; Sheridan, 2019; Triberti et al., 2020). The research in this field agrees on the negative effect of risk perceptions on user trust (Araujo et al., 2020; Hampshire, 2017; Kaplan et al.,

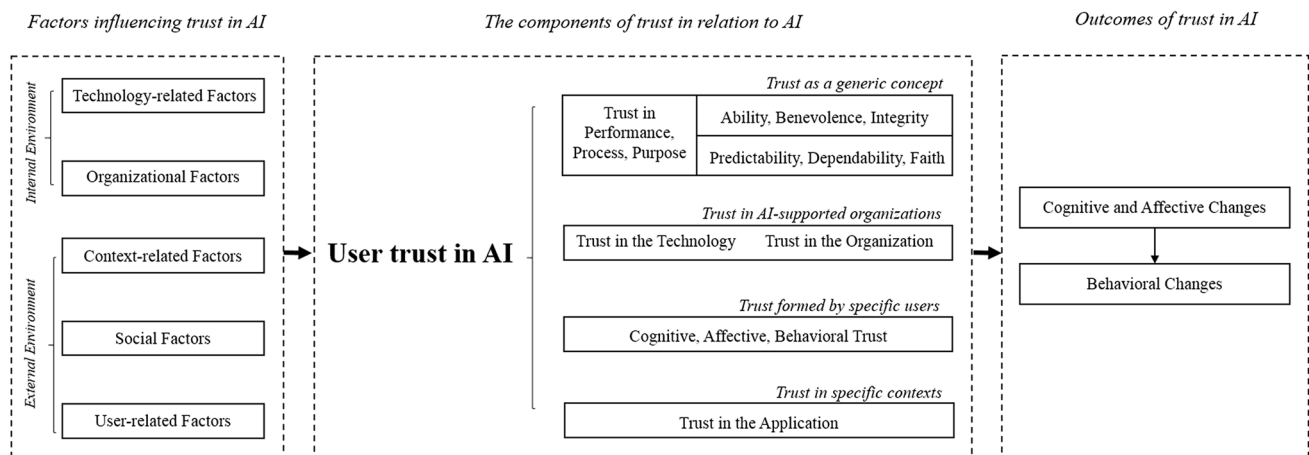


Fig. 1 The proposed framework

2021; Mou et al., 2017; Nordheim et al., 2019; Pappas, 2016; Park, 2020; Yang & Zhang, 2020; Yang et al., 2015).

Outcomes of trust in AI

Cognitive and affective changes

Trust facilitates cognitive and affective changes in AI users. In other words, trust can affect users' perceptions, opinions, beliefs, and emotions about using AI-enabled services from different providers (Vasiljeva et al., 2021). For instance, trust positively affects users' subsequent perceptions regarding the benefits of using AI (Bedué & Fritzsche, 2021). Moreover, users tend to believe using an AI-enabled system is more memorable, entertaining, exciting, comfortable, and educational after trust is built.

Attitude change is an expected outcome when a user's trust in a service is gained or lost (Kim & Peterson, 2017; Vasiljeva et al., 2021). This concept can be defined in relation to the user's disposition to respond favorably or unfavorably toward a service provider and its offers (Kim & Peterson, 2017). When AI-enabled service providers demonstrate their trustworthiness (i.e. ability, benevolence, integrity), the user's attitude and feelings are more likely to be positive, resulting in a willingness to adopt (Lee & Chen, 2022; Vasiljeva et al., 2021). Furthermore, Ameen et al. (2021) reported a positive effect of user trust on subsequent experiences. This can, in turn, lead to user satisfaction, which refers to the contentment of a user in relation to a positive experience with a product or service (Kim & Peterson, 2017). A user's trust in new technology is based on consistent and positive encounters; these also lead to a higher likelihood of satisfaction (Ashfaq et al., 2020; Hsiao & Chen, 2021; Kim & Peterson, 2017). The cognitive and affective changes may accordingly affect users' behaviors (Ashfaq et al., 2020; Bedué & Fritzsche, 2021).

Behavioral changes

Another outcome of trust is the behavioral change exhibited by the user, and this is an important domain of research. For instance, the intention to adopt or use AI has become the primary topic for studies on the consequences of user trust (Bedué & Fritzsche, 2021; Fan et al., 2020; Kim et al., 2021; Meyer-Waarden & Cloarec, 2022; Mostafa & Kasamani, 2021; Moussawi et al., 2021; Park, 2020; Priyadarshinee et al., 2017; Rahman et al., 2021; Starke et al., 2021; Vourgidis et al., 2018). The findings of these studies agree on the positive correlation between user trust and the usage of AI in various contexts. For example, Mostafa and Kasamani (2021) found that initial trust increases chatbot usage intention and user engagement in the context of online customer services. Chatterjee and Sreenivasulu (2021) reported that user trust has a positive effect on personal information-sharing behaviors on online platforms. Likewise, Park (2020) reported a positive and significant effect of trust on usage intention in the context of AI-enabled tourism services. Furthermore, users are more likely to reveal their personal and confidential information to a trusted AI (Ameen et al., 2021; Rahman et al., 2021).

However, Moussawi et al. (2021) reported a positive but insignificant effect of initial trust (i.e. ability, integrity, and benevolence) on people's intention to adopt intelligent personal agents. Notably, this study and the ones mentioned above explored the role of initial trust at an early stage of AI usage (i.e. intention to adopt), and did not consider the continuing effects of trust on users' long-term behavior. This can, however, be explained by Morgan and Hunt's (1994) commitment-trust theory, which proposes a positive effect of trust on one's relationship commitment to a product or service provider (Ameen et al., 2021). Further, Hsiao and Chen (2021), Lv et al. (2022), and Jiang and Lau (2021) suggested that user trust can affect continuance intention,

especially when fewer physical and psychological risks are perceived. User trust does not only increase the initial intention to use relevant applications but contributes to consumers' repurchase intentions and loyalty in the context of e-commerce (Bilgihan, 2016; Bitkina et al., 2020; Cheng & Jiang, 2021; Elbeltagi & Agag, 2016; Hasan et al., 2021; Kim & Peterson, 2017) and offline shopping (Ameen et al., 2021). Thus, a higher level of trust indicates that users are more committed to the service provider and less likely to switch to alternatives.

Discussions and research model

Every individual holds a general trust in other people and things along with varying trusting beliefs in different technologies. The extant literature is focused on the various aspects of users' trust in AI, including its components, influencing factors, and outcomes. The components of users' trust can be studied using different approaches. First, trust is dynamic and formed via different stages: The formation of user trust in AI starts when the user believes that the technology has shown consistent performance and is predictable. Further trust gets built on the user's overall belief in the AI's dependability in the long term. As the final stage of trust, the user has faith in the AI and will rely on the technology in the future when needed. Second, users may use different criteria to assess the trustworthiness of AI. The multiple-criteria approach suggests that trust is only formed when AI has the ability to competently perform the given tasks, present the benevolence to be aligned with users, and demonstrate the integrity to consistently follow moral principles. Some existing studies have synthesized the different stages and criteria of trust formation, namely trust in the performance, process, and purpose of AI. Furthermore, users may perceive AI as a technology-supported service or product that an organization provides. Thus, researchers have also investigated users' trust in AI in terms of their trust in the technology and service provider organizations. Moreover, the components of user trust in AI can be distinguished in terms of how they are oriented, including the associated rational cognitions, affective feelings and attitudes, and behavioral interactions. Further, the concept users' trust in AI should also consider its specific applications.

Considering the various components of user trust in AI, its influencing factors should be studied from different perspectives. The literature review findings suggest that user trust in AI is influenced by technological, organizational, contextual, social, and personal factors. First, technology-related factors concern the management of the service quality, information quality, system quality, and design quality of AI. Second, as discussed, user trust in an AI depends on components related to both the technology and the service provider organization. Thus, the organizational antecedents concern

the characteristics of AI-enabled service providers. Third, users' perceptions and assessments of the performance of AI technologies are related to the contexts of use. Therefore, the level of user trust in an AI will depend on its utilitarian and hedonic value in different contexts. Fourth, social norms and national cultures influence people's preferences, perceptions, and behaviors. Thus, they are often studied as social-related factors that affect users' trust in AI. Last, users have different prior experiences and familiarity with the technologies, personalities, and existing perceptions related to AI. Thus, the formation of trust in AI is also influenced by user-related factors.

Regarding the outcomes, users' trust affects cognitive and affective changes. Trust contributes to users' positive perceptions, opinions, beliefs, and emotions about using AI-enabled services from different providers. It can also change users' behaviors toward AI, including their intention to adopt and commitment. Based on the above discussion, Fig. 1 depicts a comprehensive conceptual framework that illustrates the components, influencing factors, and outcomes of user trust in AI.

Limitations and future research

The findings of the literature review show that the driving factors of user trust in AI can be categorized into five facets: technological, contextual, organizational, social, and user-related. It should be noted that the different facets of antecedents of user trust in AI are correlated and interact with each other. First, user perceptions of the technological qualities of AI vary in different groups. For instance, anthropomorphic designs are essential for older user groups (Rheu et al., 2021) and should also consider the personality match between the user and AI (Yen & Chiang, 2021). Such scenarios present interactions among technological and user-related factors.

Second, users' perceptions of AI, such as perceived risks, can affect their trust and acceptance levels. However, the effects of such perceived risks are moderated by the severity of consequences of risky behaviors in different contexts (Starke et al., 2021; Vassilikopoulou et al., 2018). Thus, it suggests a relationship between user-related and context-related factors.

Third, the effect of an AI's anthropomorphic facial expressions on user trust was found to be inconsistent in different experimental contexts (Rheu et al., 2021). The authors suggest that users tend to trust AI agents with male appearances when choosing utilitarian products, while they prefer recommendations from female AI agents in the context of hedonic products. Therefore, it is recommended that future studies compare the technological and user-related antecedents of user trust in AI across various contexts of use.

Fourth, a user's propensity to trust is influenced by the national culture, such as collectivism and long-term

orientation (Hallikainen & Laukkanen, 2018). These social antecedents of users' trust in AI are also linked to the technological and organizational facets. For instance, users from uncertainty-avoiding culture groups will attach more importance to the security and simplicity of new technology and its applications (Faisal et al., 2016). Service providers should be compliant with the principles of economic and social justice (Alhazmi, 2019). Thus, societal expectations, norms, and values affect the policies, principles, and procedures (e.g. data governance) of organizations that adopt AI technologies (Janssen et al., 2020). For example, more rigorous data governance is required by organizations that provide AI-enabled services to groups that are sensitive to privacy issues (Chatterjee & Sreenivasulu, 2021; Janssen et al., 2020). Despite the interactions between the different influencing factors of user trust in AI, there remains a lack of studies that simultaneously include the various facets in their research design. This, therefore, calls for more comprehensive studies in this field.

A few researchers use the term initial trust in their studies on the user-AI interactions (Fan et al., 2020; Hafizoğlu & Sen, 2018; Mostafa & Kasamani, 2021; Moussawi et al., 2021). This implies that the findings of current studies in this field tend to be limited to the early stage of trust formation. This can be attributed to several reasons: First, many studies have focused on new users' intentions to accept new AI technologies (Mostafa & Kasamani, 2021; Moussawi et al., 2021; Park, 2020; Shi et al., 2021). Second, empirical studies in this field tend to rely on data collected via cross-sectional surveys. Third, people's trust in AI has been found to build up more slowly and diminish faster than trust in human beings (Hafizoğlu & Sen, 2018; Schmidt et al., 2020). However, trust is a dynamic concept that can have long-term effects on user behaviors such as repurchase/reuse intentions, commitment, and loyalty (Ameen et al., 2021; Bilgihan, 2016; Bitkina et al., 2020; Cheng & Jiang, 2021; Elbeltagi & Agag, 2016; Kim & Perterson, 2017; Rheu et al., 2021). In practice, information technologies have been widely used to manage the total worth of a customer's business throughout the relationship period (i.e. customer lifetime value) (Bilgihan, 2016). Therefore, although users' initial trust can be significant for the early promotional stage of AI applications, more longitudinal studies are required to determine the effect of user trust in AI on the customer lifetime value.

From the commercial point of view, organizations attach importance to user trust in AI as it contributes to the sales of related products and services. Thus, the purchase and repurchase intention was a focus of the recent studies on the outcomes of user trust in AI (Rai, 2020; Yen & Chiang, 2021). However, the value of a customer for an organization should be based not only on the purchase-related aspects but also on other engagements such as providing new product ideas or referral of new customers (Barari et al., 2021). Thus, customer/user engagement is a significant concept in evaluating consumers of products and

services from both transactional and non-transactional perspectives. Nevertheless, there is still a lack of research on this topic in the context of human-AI trust. Therefore, customer/user engagement is another important topic for future research on user trust in AI, considering its relation to trust and its consequences, such as commitment (Barari et al., 2021). Furthermore, some researchers, such as Mostafa & Kasamani (2021), have discussed customer engagement as a one-dimensional concept and identified it as one of the outcomes of initial trust in AI applications. However, customer/user engagement is a multifaceted concept that, similar to trust, includes cognitive, affective, and behavioral aspects (Barari et al., 2021). This, therefore, suggests the need for more comprehensive studies that explore and compare the interactions between AI users' trust and engagement from the above-mentioned multiple perspectives.

Although our literature review identifies perceived risk as an antecedent of user trust in AI, some studies have reported a reversed relationship between these two constructs, specifically that the attenuated perceived risk is one of the outcomes of building user trust (Bedué & Fritzsche, 2021; Gillath et al., 2021; Mou et al., 2017; Sutrop, 2019). This has led to confusion about the actual relationship between user trust and perceived risk. However, a few studies suggested perceived risk as an outcome of user trust that presents inconsistent model fits for their proposed theoretical frameworks (Mou et al., 2017). Notably, these studies tend to consider perceived risk and uncertainty as the same concept (Bedué & Fritzsche, 2021); however, these two constructs should be distinguished (Yang & Wibowo, 2022). People perceive risks when they are able to clearly identify and evaluate the outcomes and probability of an unfavorable event. In contrast, perceived uncertainties indicate the ambiguity of future events, with people being unable to make precise predictions (Yang & Wibowo, 2020). As discussed, users' trust is based on their confidence in the performance, process, and purpose of AI applications. This confidence should be based on the users' ability to identify and evaluate risks (i.e. perceived risk as an antecedent of trust). Further, consumers tend to rely on their trust in the service provider to reduce the uncertainty in decision making when lacking other precise references (i.e. perceived uncertainty as the outcome of trust) (Yang et al., 2019). Therefore, it is necessary to distinguish the roles of perceived risk and uncertainty in future research on user trust in AI.

The current study explores the various factors affecting users' trust in AI. These factors suggest that the influencing factors in trust building can be studied at the macro (e.g. national culture) and micro (e.g. users' personality) levels. This approach is consistent with previous research on user trust in other relevant fields, e.g. e-business (Kim & Tadisina, 2007) and public services (Wilkes, 2014). Further, we proposed users' trust in AI as a concept that includes multiple components. However, the existing literature on the various influencing factors tends to

consider users' trust as a one-dimensional construct. For example, previous studies on AI have demonstrated that users' trust can be affected by factors related to the technology (Nasirian et al., 2017; Pitardi & Marriott, 2021; Qin et al., 2020), organization (Nordheim et al., 2019), context (Bilgihan 2016; Qin et al., 2020), culture (Robinson, 2020), and individual differences (Qin et al., 2020; Sheridan, 2019); nevertheless, these studies only discussed "trust" as a generic concept. Accordingly, this study suggested that an influencing factor may affect the multiple components of users' trust in AI. For instance, individual differences are related to users' generic trust (Sheridan, 2019). They also affect users' cognitive, affective and behavioral trust (Khaksari & Keyvanpour, 2019), and the trust in the technology, organization (Łapińska et al., 2021) and specific applications (Oksanen et al., 2020; Qin et al., 2020). However, it still requires future studies to investigate whether other influencing factors can consistently affect the various components of users' trust in AI.

Further, the comparison among different components of trust that can affect users' perceptions and behaviors toward AI is also under-researched. For instance, Bedué and Fritzsche (2021) argue that trust can simultaneously affect users' perceived benefits and intention to use AI. However, trust was only discussed as a generic concept reflected by ability, integrity and benevolence in their study. Although the study considered the applications of AI in various sectors, it did not clarify whether users' trust in the organization and the technology similarly play similar or different roles in the change of perceptions and behaviors. By comparison, Li et al. (2021) distinguished the trust in the organization and the technology. Nevertheless, the study did not explore the outcome of building users' trust based on the two distinctive aspects. Although existing research in this field tends to agree on users' cognitive, affective, and behavioral changes as the outcome of trust building (Bedué & Fritzsche, 2021; Fan et al., 2020; Kim et al., 2021; Meyer-Waarden & Cloarec, 2022; Mostafa & Kasamani, 2021; Moussawi et al., 2021; Park, 2020; Priyadarshinee et al., 2017; Rahman et al., 2021; Starke et al., 2021; Vasiljeva et al., 2021; Vourgidis et al., 2018), future studies should further test and compare the roles of various components of users trust. Furthermore, from the users' decision-making perspective, how people trust and accept AI depends on their involvement with AI applications (Mostafa & Kasamani, 2021). In other words, people will use simpler decision rules when they believe some AI applications are less valuable or important (Bauer et al., 2006; Hansen, 2005). Thus, more empirical studies should be conducted to investigate whether all the components of trust are required for people's acceptance of AI applications with different user involvement levels.

Last, this study was conducted from the user perspective. However, the interaction among users, organizations and society should not be ignored. Building users' trust can contribute to the digitalization of organizations in different ways, including encouraging new product developments, and reducing

transaction costs and investment risks (Jelovac et al., 2021; Weber-Lewerenz, 2021). Therefore, it suggests more empirical studies to investigate the long-term impact of user trust on AI-powered organizations. Further, the "black-box" element of AI is still one of the important issues in this field because "the machine learning systems used for predicting an outcome remain opaque or cannot be adequately observed" (Elliott et al., 2021, p.180). Accordingly, the adoption of AI may cause unanticipated or unintended consequences that can either positively or negatively impact society in the long term (Pringle et al., 2016). For example, a higher level of user trust may be associated with a higher possibility of misuse of AI. Thus, further research in this area is also necessary.

Conclusion

The findings of this review provide insights into the field of user trust in AI. The review provides a detailed research outlook of the given area, including the components, influencing factors, and outcomes of user trust in AI. Based on the findings, a new comprehensive and conceptual framework was proposed for a better understanding of user trust in AI. From a theoretical perspective, the proposed framework builds on existing theories and provides potential future research avenues. Building on existing theoretical models (e.g. UTAUT and CDR), this framework presents a logical sequence linking the influences, components and outcomes, from a new perspective that is more related to AI trust-building. Thus, this study more extensively explores the key factors in relation to user trust in AI, and suggests the interactions among relevant constructs. However, the framework needs to be tested and validated in more specific contexts (e.g. collaborative filtering, content-based filtering, intelligent systems). From a practical perspective, this study helps AI-supported service providers understand the concept of user trust from various perspectives, and emphasizes the need to build trust based on different facets (i.e. technological, organizational, contextual, social, and user-related) to facilitate positive cognitive, affective, and behavioral changes among the users. Therefore, this review provides a foundation for future researchers and practitioners to obtain more comprehensive findings.

References

- Agag, G. M., & El-Masry, A. A. (2017). Why do consumers trust online travel websites? Drivers and outcomes of consumer trust toward online travel websites. *Journal of Travel Research*, 56(3), 347–369. <https://doi.org/10.1177/0047287516643185>
- Ahmadi, V., & Tutschku, K. (2017). Privacy and trust in cloud-based marketplaces for AI and data resources. *Proceedings of the 11th IFIP International Conference on Trust Management*. Gothenburg, Sweden. 223–225.
- Ahn, J., Kim, J., & Sung, Y. (2022). The effect of gender stereotypes on artificial intelligence recommendations. *Journal of Business*

- Research, 141, 50–59. <https://doi.org/10.1016/j.jbusres.2021.12.007>
- Ajzen, I. (1991). The theory of planned behavior. *Organizational Behavior and Human Decision Processes*, 50, 179–211. [https://doi.org/10.1016/0749-5978\(91\)90020-T](https://doi.org/10.1016/0749-5978(91)90020-T)
- Alaieri, F., & Vellino, A. (2016). Ethical decision making in robots: Autonomy, trust and responsibility. *Proceedings of the International Conference on Social Robotics.*, 159–168. https://doi.org/10.1007/978-3-319-47437-3_16
- Alhazmi, B. M. (2019). Religiosity and customer trust in financial services marketing relationships. *Journal of Financial Services Marketing*, 24(1), 31–43. <https://doi.org/10.1057/s41264-019-00062-9>
- Ameen, N., Tarhini, A., Reppel, A., & Anand, A. (2021). Customer experiences in the age of artificial intelligence. *Computers in Human Behavior*, 114, 106548. <https://doi.org/10.1016/j.chb.2020.106548>
- Anwar, M. (2020). Supporting privacy, trust, and personalization in online learning. *International Journal of Artificial Intelligence in Education*, 31, 1–15. <https://doi.org/10.1007/s40593-020-00216-0>
- Araujo, T., Helberger, N., Kruijemeier, S., & De Vreese, C. H. (2020). In AI we trust? Perceptions about automated decision-making by artificial intelligence. *AI & SOCIETY*, 35(3), 611–623. <https://doi.org/10.1007/s00146-019-00931-w>
- Ashfaq, M., Yun, J., Yu, S., & Loureiro, S. M. C. (2020). I. Chatbot: Modeling the determinants of users' satisfaction and continuance intention of AI-powered service agents. *Telematics and Informatics*, 54, 101473. <https://doi.org/10.1016/j.tele.2020.101473>
- Balakrishnan, J., & Dwivedi, Y. K. (2021). Role of cognitive absorption in building user trust and experience. *Psychology & Marketing*, 38(4), 643–668. <https://doi.org/10.1002/mar.21462>
- Banks, J. (2020). Good robots, bad robots: Morally valenced behavior effects on perceived mind, morality, and trust. *International Journal of Social Robotics*, 13, 2021–2038. <https://doi.org/10.1007/s12369-020-00692-3>
- Barari, M., Ross, M., Thaichon, S., & Surachartkumtonkun, J. (2021). A meta-analysis of customer engagement behaviour. *International Journal of Consumer Studies*, 45(4), 457–477. <https://doi.org/10.1111/ijcs.12609>
- Bauer, H. H., Sauer, N. E., & Becker, C. (2006). Investigating the relationship between product involvement and consumer decision-making styles. *Journal of Consumer Behaviour*, 5(4), 342–354. <https://doi.org/10.1002/cb.185>
- Baum, Z. J., Yu, X., Ayala, P. Y., Zhao, Y., Watkins, S. P., & Zhou, Q. (2021). Artificial intelligence in chemistry: Current trends and future directions. *Journal of Chemical Information and Modeling*, 61(7), 3197–3212. <https://doi.org/10.1021/acs.jcim.1c00619>
- Bawack, R. E., Wamba, S. F., & Carillo, K. D. A. (2021). Exploring the role of personality, trust, and privacy in customer experience performance during voice shopping: Evidence from SEM and fuzzy set qualitative comparative analysis. *International Journal of Information Management*, 58, 102309. <https://doi.org/10.1016/j.ijinfomgt.2021.102309>
- Bedué, P., & Fritzsche, A. (2021). Can we trust AI? An empirical investigation of trust requirements and guide to successful AI adoption. *Journal of Enterprise Information Management.*, 35(2), 530–549. <https://doi.org/10.1108/JEIM-06-2020-0233>
- Bejger, S., & Elster, S. (2020). Artificial intelligence in economic decision making: How to assure a trust? *Ekonomia i Prawo. Economics and Law*, 19(3), 411–434. <https://doi.org/10.12775/EiP.2020.028>
- Bernotat, J., Eyssel, F., & Sachse, J. (2017). Shape it—the influence of robot body shape on gender perception in robots. *Proceedings of the International Conference on Social Robotics.*, 75–84. https://doi.org/10.1007/978-3-319-70022-9_8
- Bilgihan, A. (2016). Gen Y customer loyalty in online shopping: An integrated model of trust, user experience and branding. *Computers in Human Behavior*, 61, 103–113. <https://doi.org/10.1016/j.chb.2016.03.014>
- Bitkina, O. V., Jeong, H., Lee, B. C., Park, J., Park, J., & Kim, H. K. (2020). Perceived trust in artificial intelligence technologies: A preliminary study. *Human Factors and Ergonomics in Manufacturing & Service Industries*, 30(4), 282–290. <https://doi.org/10.1002/hfm.20839>
- Burda, D., & Teuteberg, F. (2014). The role of trust and risk perceptions in cloud archiving—Results from an empirical study. *The Journal of High Technology Management Research*, 25(2), 172–187. <https://doi.org/10.1016/j.hitech.2014.07.008>
- Chatterjee, S., & Sreenivasulu, N. (2021). Impact of AI regulation and governance on online personal data sharing: From socio-legal, technology and policy perspective. *Journal of Science and Technology Policy Management*. <https://doi.org/10.1108/JSTPM-07-2020-0103>
- Chen, Y. K., & Wen, C. R. (2021). Impacts of attitudes toward government and corporations on public trust in artificial intelligence. *Communication Studies*, 72(1), 115–131. <https://doi.org/10.1080/10510974.2020.1807380>
- Cheng, Y., & Jiang, H. (2021). Customer–brand relationship in the era of artificial intelligence: Understanding the role of chatbot marketing efforts. *Journal of Product & Brand Management.*, 31(2), 252–264. <https://doi.org/10.1108/JPBPM-05-2020-2907>
- Chong, L., Zhang, G., Goucher-Lambert, K., Kotovsky, K., & Cagan, J. (2022). Human confidence in artificial intelligence and in themselves: The evolution and impact of confidence on adoption of AI advice. *Computers in Human Behavior*, 127, 107018. <https://doi.org/10.1016/j.chb.2021.107018>
- Crigger, E., Reinbold, K., Hanson, C., Kao, A., Blake, K., & Irons, M. (2022). Trustworthy augmented intelligence in health care. *Journal of Medical Systems*, 46(2), 1–11. <https://doi.org/10.1007/s10916-021-01790-z>
- Crockett, K., Garratt, M., Latham, A., Colyer, E., & Goltz, S. (2020). Risk and trust perceptions of the public of artificial intelligence applications. In *Proceedings of the 2020 International Joint Conference on Neural Networks*. <https://doi.org/10.1109/IJCNN48605.2020.9207654>
- Davis, F. D. (1985). *A technology acceptance model for empirically testing new end-user information systems: Theory and results*. Doctoral dissertation, Massachusetts Institute of Technology.
- Dos Santos, D. P., Giese, D., Brodehl, S., Chon, S., Staab, W., Kleinert, R., & Baeßler, B. (2019). Medical students' attitude towards artificial intelligence: A multicentre survey. *European Radiology*, 29(4), 1640–1646. <https://doi.org/10.1007/s00330-018-5601-1>
- Dreyer, S., Olivotti, D., Lebek, B., & Breitner, M. H. (2019). Focusing the customer through smart services: A literature review. *Electronic Markets*, 29(1), 55–78. <https://doi.org/10.1007/s12525-019-00328-z>
- Elbeltagi, I., & Agag, G. (2016). E-retailing ethics and its impact on customer satisfaction and repurchase intention: A cultural and commitment-trust theory perspective. *Internet Research*, 26(1), 288–310. <https://doi.org/10.1108/IntR-10-2014-0244>
- Elliott, K., Price, R., Shaw, P., Spiliotopoulos, T., Ng, M., Coopamootoo, K., & van Moorsel, A. (2021). Towards an equitable digital society: Artificial intelligence (AI) and corporate digital responsibility (CDR). *Society*, 58(3), 179–188. <https://doi.org/10.1007/s12115-021-00594-8>
- Eubanks, V. (2018). *Automating inequality: How high-tech tools profile, police, and punish the poor*. Martin's Press.
- Everson, C. (2003). *Organizational environment* (4th ed.). Pergamon Flexible Learning. <https://doi.org/10.4324/9780080497600>

- Faisal, C. N., Gonzalez-Rodriguez, M., Fernandez-Lanvin, D., & de Andres-Suarez, J. (2016). Web design attributes in building user trust, satisfaction, and loyalty for a high uncertainty avoidance culture. *IEEE Transactions on Human-Machine Systems*, 47(6), 847–859. <https://doi.org/10.1109/THMS.2016.2620901>
- Fan, W., Liu, J., Zhu, S., & Pardalos, P. M. (2020). Investigating the impacting factors for the healthcare professionals to adopt artificial intelligence-based medical diagnosis support system (AIMDSS). *Annals of Operations Research*, 294(1), 567–592. <https://doi.org/10.1007/s10479-018-2818-y>
- Fang, H., Li, X., & Zhang, J. (2022). Integrating social influence modeling and user modeling for trust prediction in signed networks. *Artificial Intelligence*, 302, 103628. <https://doi.org/10.1016/j.artint.2021.103628>
- Fatima, S., Desouza, K. C., Denford, J. S., & Dawson, G. S. (2021). What explains governments interest in artificial intelligence? A signaling theory approach. *Economic analysis and policy*, 71, 238–254. <https://doi.org/10.1016/j.eap.2021.05.001>
- Felzmann, H., Villaronga, E. F., Lutz, C., & Tamò-Larrieux, A. (2019). Transparency you can trust: Transparency requirements for artificial intelligence between legal norms and contextual concerns. *Big Data & Society*, 6(1), 2053951719860542. <https://doi.org/10.1177/2053951719860542>
- Fishbein, M., & Ajzen, I. (1977). Belief, attitude, intention, and behavior: An introduction to theory and research. *Philosophy and Rhetoric*, 10(2), 177–188.
- Følstad, A., Nordheim, C. B., & Bjørkli, C. A. (2018). What makes users trust a chatbot for customer service? An exploratory interview study. *Proceedings of the International Conference on Internet Science*. https://doi.org/10.1007/978-3-030-01437-7_16
- Gansser, O. A., & Reich, C. S. (2021). A new acceptance model for artificial intelligence with extensions to UTAUT2: An empirical study in three segments of application. *Technology in Society*, 65, 101535. <https://doi.org/10.1016/j.techsoc.2021.101535>
- Giampietri, E., Verneau, F., Del Giudice, T., Carfora, V., & Finco, A. (2018). A theory of planned behaviour perspective for investigating the role of trust in consumer purchasing decision related to short food supply chains. *Food Quality and Preference*, 64, 160–166. <https://doi.org/10.1016/j.foodqual.2017.09.012>
- Gillath, O., Ai, T., Branicky, M. S., Keshmiri, S., Davison, R. B., & Spaulding, R. (2021). Attachment and trust in artificial intelligence. *Computers in Human Behavior*, 115, 106607. <https://doi.org/10.1016/j.chb.2020.106607>
- Girsang, M. J., Hendayani, R., & Ganesan, Y. (2020). Can information security, privacy and satisfaction influence the e-commerce consumer trust? *Proceedings of the 8th International Conference on Information and Communication Technology*. <https://doi.org/10.1109/ICoICT49345.2020.9166247>
- Glikson, E., & Woolley, A. W. (2020). Human trust in artificial intelligence: Review of empirical research. *Academy of Management Annals*, 14(2), 627–660. <https://doi.org/10.5465/annals.2018.0057>
- Gulati, S., Sousa, S., & Lamas, D. (2019). Design, development and evaluation of a human-computer trust scale. *Behaviour & Information Technology*, 38(10), 1004–1015. <https://doi.org/10.1080/0144929X.2019.1656779>
- Gupta, S., Kamboj, S., & Bag, S. (2021). Role of risks in the development of responsible artificial intelligence in the digital healthcare domain. *Information Systems Frontiers*, 1–18. <https://doi.org/10.1007/s10796-021-10174-0>
- Hafizoğlu, F. M., & Sen, S. (2018). The effects of past experience on trust in repeated human-agent teamwork. In *Proceedings of the 17th International Conference on Autonomous Agents and Multi Agent Systems, Stockholm, Sweden*.
- Hallikainen, H., & Laukkanen, T. (2018). National culture and consumer trust in e-commerce. *International Journal of Information Management*, 38(1), 97–106. <https://doi.org/10.1016/j.ijinfomgt.2017.07.002>
- Hamid, N. H. A., Ahmad, M. S., Ahmad, A., Mustapha, A., Mahmoud, M. A., & Yusoff, M. Z. M. (2015). Trusting norms: A conceptual norms' trust framework for norms adoption in open normative multi-agent systems. *Proceedings of the 12th International Conference on Distributed Computing and Artificial Intelligence*. https://doi.org/10.1007/978-3-319-19638-1_17
- Hampshire, C. (2017). A mixed methods empirical exploration of UK consumer perceptions of trust, risk and usefulness of mobile payments. *International Journal of Bank Marketing*, 35(3), 354–369. <https://doi.org/10.1108/IJBM-08-2016-0105>
- Hansen, T. (2005). Perspectives on consumer decision making: An integrated approach. *Journal of Consumer Behaviour*, 4(6), 420–437. <https://doi.org/10.1002/cb.33>
- Hasan, R., Shams, R., & Rahman, M. (2021). Consumer trust and perceived risk for voice-controlled artificial intelligence: The case of Siri. *Journal of Business Research*, 131, 591–597. <https://doi.org/10.1016/j.jbusres.2020.12.012>
- Hepworth, A. J., Baxter, D. P., Hussein, A., Yaxley, K. J., Debie, E., & Abbass, H. A. (2020). Human-swarm-teaming transparency and trust architecture. *IEEE/CAA Journal of Automatica Sinica*, 8(7), 1281–1295. <https://doi.org/10.1109/JAS.2020.1003545>
- Hong, J. W., & Williams, D. (2019). Racism, responsibility and autonomy in HCI: Testing perceptions of an AI agent. *Computers in Human Behavior*, 100, 79–84. <https://doi.org/10.1016/j.chb.2019.06.012>
- Hsiao, K. L., & Chen, C. C. (2021). What drives continuance intention to use a food-ordering chatbot? An examination of trust and satisfaction. *Library Hi Tech*. Advance online publication. <https://doi.org/10.1108/LHT-08-2021-0274>
- Hua, C., Cole, S., & Xu, N. (2021). Rethinking trust in tourism apps: The moderating effect of age. *Journal of Hospitality and Tourism Technology*, 12(3), 548–562. <https://doi.org/10.1108/JHTT-01-2020-0013>
- Jelovac, D., Ljubojević, Č., & Ljubojević, L. (2021). HPC in business: The impact of corporate digital responsibility on building digital trust and responsible corporate digital governance. *Digital Policy, Regulation and Governance*. <https://doi.org/10.1108/DPRG-11-2020-0164>
- Jain, S., Luthra, M., Sharma, S., & Fatima, M. (2020). Trustworthiness of artificial intelligence. *Proceedings of the 2020 6th International Conference on Advanced Computing and Communication Systems*. <https://doi.org/10.1109/ICACCS48705.2020.9074237>
- Janssen, M., Brous, P., Estevez, E., Barbosa, L. S., & Janowski, T. (2020). Data governance: Organizing data for trustworthy artificial intelligence. *Government Information Quarterly*, 37(3), 101493. <https://doi.org/10.1016/j.giq.2020.101493>
- Jiang, Y., & Lau, A. K. (2021). Roles of consumer trust and risks on continuance intention in the sharing economy: An empirical investigation. *Electronic Commerce Research and Applications*, 47, 101050. <https://doi.org/10.1016/j.elerap.2021.101050>
- Jones, V. (2018). Voice-activated change: Marketing in the age of artificial intelligence and virtual assistants. *Journal of Brand Strategy*, 7(3), 233–245.
- Jung, S. (2015). Does education affect risk aversion? Evidence from the British education reform. *Applied Economics*, 47(28), 2924–2938. <https://doi.org/10.1080/00036846.2015.1011313>
- Kaplan, A. D., Kessler, T. T., Brill, J. C., & Hancock, P. (2021). Trust in artificial intelligence: Meta-analytic findings. *Human Factors*, 00187208211013988. <https://doi.org/10.1177/00187208211013988>
- Kerasidou, A. (2020). Artificial intelligence and the ongoing need for empathy, compassion and trust in healthcare. *Bulletin of the World Health Organization*, 98(4), 245. <https://doi.org/10.2471/BLT.19.237198>
- Khaksari, A., & Keyvanpour, M. (2019). TP-TA: A comparative analytical framework for trust prediction models in online social networks based on trust aspects. *Artificial Intelligence Review*, 52(3), 1929–1960. <https://doi.org/10.1007/s10462-017-9583-1>

- Kim, Y., & Peterson, R. A. (2017). A meta-analysis of online trust relationships in e-commerce. *Journal of Interactive Marketing*, 38, 44–54. <https://doi.org/10.1016/j.intmar.2017.01.001>
- Kim, T., & Song, H. (2021). How should intelligent agents apologize to restore trust? Interaction effects between anthropomorphism and apology attribution on trust repair. *Telematics and Informatics*, 61, 101595. <https://doi.org/10.1016/j.tele.2021.101595>
- Kim, E., & Tadisina, S. (2007). A model of customers' trust in e-businesses: micro-level inter-party trust formation. *Journal of Computer Information Systems*, 48(1), 88–104.
- Kim, J., Giroux, M., & Lee, J. C. (2021). When do you trust AI? The effect of number presentation detail on consumer trust and acceptance of AI recommendations. *Psychology & Marketing*, 39(2), 370–389. <https://doi.org/10.1002/mar.21498>
- Kolasinska, A., Lauriola, I., & Quadrio, G. (2019). Do people believe in artificial intelligence? A cross-topic multicultural study. *Proceedings of the 5th EAI International Conference on Smart Objects and Technologies for Social Good*, 31–36. <https://doi.org/10.1145/3342428.3342667>
- Kraus, M., Wagner, N., Callejas, Z., & Minker, W. (2021). The role of trust in proactive conversational assistants. *IEEE Access*, 9, 112821–112836. <https://doi.org/10.1109/ACCESS.2021.3103893>
- Kuipers, B. (2018). How can we trust a robot? *Communications of the ACM*, 61(3), 86–95. <https://doi.org/10.1145/3173087>
- Landwehr, C. (2019). 2018: A big year for privacy. *Communications of the ACM*, 62(2), 20–22. <https://doi.org/10.1145/3300224>
- Langer, A., Feingold-Polak, R., Mueller, O., Kellmeyer, P., & Levy-Tzedek, S. (2019). Trust in socially assistive robots: Considerations for use in rehabilitation. *Neuroscience & Biobehavioral Reviews*, 104, 231–239. <https://doi.org/10.1016/j.neubiorev.2019.07.014>
- Łapińska, J., Escher, I., Górka, J., Sudolska, A., & Brzustewicz, P. (2021). Employees' trust in artificial intelligence in companies: The case of energy and chemical industries in Poland. *Energies*, 14(7), 1942. <https://doi.org/10.3390/en14071942>
- Lazanyi, K. (2018). Readiness for artificial intelligence. *Proceedings of the 2018 IEEE 16th International Symposium on Intelligent Systems and Informatics*. <https://doi.org/10.1109/SISY.2018.8524740>
- Lee, J. C., & Chen, X. (2022). Exploring users' adoption intentions in the evolution of artificial intelligence mobile banking applications: The intelligent and anthropomorphic perspectives. *International Journal of Bank Marketing*, 40(4), 631–658. <https://doi.org/10.1108/IJBM-08-2021-0394>
- Lei, X., & Rau, P. L. P. (2021). Effect of relative status on responsibility attributions in human–robot collaboration: Mediating role of sense of responsibility and moderating role of power distance orientation. *Computers in Human Behavior*, 122, 106820. <https://doi.org/10.1016/j.chb.2021.106820>
- Li, J., Sun, C., & Lv, J. (2014). TCMF: Trust-based context-aware matrix factorization for collaborative filtering. In *26th International Conference on Tools with Artificial Intelligence* (pp. 815–821). <https://doi.org/10.1109/ICTAI.2014.126>
- Li, J., Zhou, Y., Yao, J., & Liu, X. (2021). An empirical investigation of trust in AI in a Chinese petrochemical enterprise based on institutional theory. *Scientific Reports*, 11(1), 1–12. <https://doi.org/10.1038/s41598-021-92904-7>
- Liao, X., Ding, L., & Wang, Y. (2011). Secure machine learning, a brief overview. In *5th International Conference on Secure Software Integration and Reliability Improvement-Companion* (pp. 26–29). <https://doi.org/10.1109/SSIRI-C.2011.15>
- Liyanaarachchi, G., Deshpande, S., & Weaven, S. (2020). Market-oriented corporate digital responsibility to manage data vulnerability in online banking. *International Journal of Bank Marketing*, 39(4), 571–591. <https://doi.org/10.1108/IJBM-06-2020-0313>
- Lobschat, L., Mueller, B., Eggers, F., Brandimarte, L., Diefenbach, S., Kroschke, M., & Wirtz, J. (2021). Corporate digital responsibility. *Journal of Business Research*, 122, 875–888. <https://doi.org/10.1016/j.jbusres.2019.10.006>
- Lou, C., & Yuan, S. (2019). Influencer marketing: How message value and credibility affect consumer trust of branded content on social media. *Journal of Interactive Advertising*, 19(1), 58–73. <https://doi.org/10.1080/15252019.2018.1533501>
- Lu, L., McDonald, C., Kelleher, T., Lee, S., Chung, Y. J., Mueller, S., & Yue, C. A. (2022). Measuring consumer-perceived humanness of online organizational agents. *Computers in Human Behavior*, 128, 107092. <https://doi.org/10.1016/j.chb.2021.107092>
- Luxton, D. D. (2014). Recommendations for the ethical use and design of artificial intelligent care providers. *Artificial Intelligence in Medicine*, 62(1), 1–10. <https://doi.org/10.1016/j.artmed.2014.06.004>
- Lv, X., Yang, Y., Qin, D., Cao, X., & Xu, H. (2022). Artificial intelligence service recovery: The role of empathic response in hospitality customers' continuous usage intention. *Computers in Human Behavior*, 126, 106993. <https://doi.org/10.1016/j.chb.2021.106993>
- Mal, C. I., Davies, G., & Diers-Lawson, A. (2018). Through the looking glass: The factors that influence consumer trust and distrust in brands. *Psychology & Marketing*, 35(12), 936–947. <https://doi.org/10.1002/mar.21146>
- Mañero, J. (2020). Review of Virginia Eubanks (2018). Automating inequality: How high-tech tools profile, police, and punish the poor. *Postdigital Science and Education*, 2(2), 489–493. <https://doi.org/10.1007/s42438-019-00077-4>
- Meske, C., & Bunde, E. (2020). Transparency and trust in human-AI-interaction: The role of model-agnostic explanations in computer vision-based decision support. *Proceedings of the International Conference on Human-Computer Interaction*, 54–69. https://doi.org/10.1007/978-3-030-50334-5_4
- Meyer-Waarden, L., & Cloarec, J. (2022). Baby, you can drive my car: Psychological antecedents that drive consumers' adoption of AI-powered autonomous vehicles. *Technovation*, 109, 102348. <https://doi.org/10.1016/j.technovation.2021.102348>
- Mohanty, H. (2020). Trust: Anthropomorphic algorithmic. *Proceedings of the International Conference on Distributed Computing and Internet Technology*, 50–72. https://doi.org/10.1007/978-3-030-36987-3_4
- Morgan, R. M., & Hunt, S. D. (1994). The commitment-trust theory of relationship marketing. *Journal of Marketing*, 58(3), 20–38. <https://doi.org/10.1177/002224299405800302>
- Mostafa, R. B., & Kasamani, T. (2021). Antecedents and consequences of chatbot initial trust. *European Journal of Marketing*. <https://doi.org/10.1108/EJM-02-2020-0084>
- Mou, J., Shin, D.-H., & Cohen, J. F. (2017). Trust and risk in consumer acceptance of e-services. *Electronic Commerce Research*, 17(2), 255–288. <https://doi.org/10.1007/s10660-015-9205-4>
- Moussawi, S., Koufaris, M., & Benbunan-Fich, R. (2021). How perceptions of intelligence and anthropomorphism affect adoption of personal intelligent agents. *Electronic Markets*, 31(2), 343–364. <https://doi.org/10.1007/s12525-020-00411-w>
- Nasirian, F., Ahmadian, M., & Lee, O. K. D. (2017). AI-based voice assistant systems: Evaluating from the interaction and trust perspectives. In *Proceedings of the Twenty-third Americas Conference on Information Systems, Boston, USA*.
- Nassar, M., Salah, K., ur Rehman, M. H., & Svetinovic, D. (2020). Blockchain for explainable and trustworthy artificial

- intelligence. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 10(1), e1340. <https://doi.org/10.1002/widm.1340>
- Nguyen, N., & Pervan, S. (2020). Retailer corporate social responsibility and consumer citizenship behavior: The mediating roles of perceived consumer effectiveness and consumer trust. *Journal of Retailing and Consumer Services*, 55, 102082. <https://doi.org/10.1016/j.jretconser.2020.102082>
- Nilashi, M., Ibrahim, O., Mirabi, V. R., Ebrahimi, L., & Zare, M. (2015). The role of security, design and content factors on customer trust in mobile commerce. *Journal of Retailing and Consumer Services*, 26, 57–69. <https://doi.org/10.1016/j.jretconser.2015.05.002>
- Nordheim, C. B., Følstad, A., & Bjørkli, C. A. (2019). An initial model of trust in chatbots for customer service-findings from a questionnaire study. *Interacting with Computers*, 31(3), 317–335. <https://doi.org/10.1093/iwc/iwz022>
- Ntoutsis, E., Fafalios, P., Gadiraju, U., Iosifidis, V., Nejd, W., Vidal, M. E., Ruggieri, S., Turini, F., Papadopoulos, S., Krasanakis, E., Kompatsiaris, I., Kinder-Kurlanda, K., Wagner, C., Karimi, F., Fernandez, M., Alani, H., Berendt, B., Kruegel, T., Heinze, C., Broelemann, K., Kasneci, G., Tiropanis, T., & Staab, S. (2020). Bias in data-driven artificial intelligence systems - an introductory survey. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 10(3), e1356. <https://doi.org/10.1002/widm.1356>
- Nundy, S., Montgomery, T., & Wachter, R. M. (2019). Promoting trust between patients and physicians in the era of artificial intelligence. *Jama*, 322(6), 497–498. <https://doi.org/10.1001/jama.2018.20563>
- Oksanen, A., Savela, N., Latikka, R., & Koivula, A. (2020). Trust toward robots and artificial intelligence: An experimental approach to human-technology interactions online. *Frontiers in Psychology*, 11. <https://doi.org/10.3389/fpsyg.2020.568256>
- Oliveira, T., Alinho, M., Rita, P., & Dhillon, G. (2017). Modelling and testing consumer trust dimensions in e-commerce. *Computers in Human Behavior*, 71, 153–164. <https://doi.org/10.1016/j.chb.2017.01.050>
- Pappas, N. (2016). Marketing strategies, perceived risks, and consumer trust in online buying behaviour. *Journal of Retailing and Consumer Services*, 29, 92–103. <https://doi.org/10.1016/j.jretconser.2015.11.007>
- Paradedá, R. B., Hashemian, M., Rodrigues, R. A., & Paiva, A. (2016). How facial expressions and small talk may influence trust in a robot. *Proceedings of the International Conference on Social Robotics*. https://doi.org/10.1007/978-3-319-47437-3_17
- Park, S. (2020). Multifaceted trust in tourism service robots. *Annals of Tourism Research*, 81, 102888. <https://doi.org/10.1016/j.annals.2020.102888>
- Paek, S., & Kim, N. (2021). Analysis of worldwide research trends on the impact of artificial intelligence in education. *Sustainability*, 13(14), 7941. <https://doi.org/10.1016/j.annals.2020.102888>
- Pelau, C., Dabija, D.-C., & Ene, I. (2021). What makes an AI device human-like? The role of interaction quality, empathy and perceived psychological anthropomorphic characteristics in the acceptance of artificial intelligence in the service industry. *Computers in Human Behavior*, 122, 106855. <https://doi.org/10.1016/j.chb.2021.106855>
- Pena, M. L. C., Carballal, A., Rodríguez-Fernández, N., Santos, I., & Romero, J. (2021). Artificial intelligence applied to conceptual design. A review of its use in architecture. *Automation in Construction*, 124, 103550. <https://doi.org/10.1016/j.autcon.2021.103550>
- Pickering, B. (2021). Trust, but verify: Informed consent, AI technologies, and public health emergencies. *Future Internet*, 13(5), 132. <https://doi.org/10.3390/fi13050132>
- Pieters, W. (2011). Explanation and trust: What to tell the user in security and AI? *Ethics and Information Technology*, 13(1), 53–64. <https://doi.org/10.1007/s10676-010-9253-3>
- Pitardi, V., & Marriott, H. R. (2021). Alexa, she's not human but... unveiling the drivers of consumers' trust in voice-based artificial intelligence. *Psychology & Marketing*, 38(4), 626–642. <https://doi.org/10.1002/mar.21457>
- Priyadarshinee, P., Raut, R. D., Jha, M. K., & Gardas, B. B. (2017). Understanding and predicting the determinants of cloud computing adoption: A two staged hybrid SEM-neural networks approach. *Computers in Human Behavior*, 76, 341–362. <https://doi.org/10.1016/j.chb.2017.07.027>
- Pringle, R., Michael, K., & Michael, M. G. (2016). Unintended consequences of living with AI: The paradox of technological potential? *IEEE Technology and Society Magazine*, 35(4), 17–21. <https://doi.org/10.1109/MTS.2016.2632978>
- Qin, F., Li, K., & Yan, J. (2020). Understanding user trust in artificial intelligence-based educational systems: Evidence from China. *British Journal of Educational Technology*, 51(5), 1693–1710. <https://doi.org/10.1111/bjet.12994>
- Rahman, M., Ming, T. H., Baigh, T. A., & Sarker, M. (2021). Adoption of artificial intelligence in banking services: An empirical analysis. *International Journal of Emerging Markets*. <https://doi.org/10.1108/IJOEM-06-2020-0724>
- Rai, A. (2020). Explainable AI: From black box to glass box. *Journal of the Academy of Marketing Science*, 48(1), 137–141. <https://doi.org/10.1007/s11747-019-00710-5>
- Rheu, M., Shin, J. Y., Peng, W., & Huh-Yoo, J. (2021). Systematic review: Trust-building factors and implications for conversational agent design. *International Journal of Human-Computer Interaction*, 37(1), 81–96. <https://doi.org/10.1080/10447318.2020.1807710>
- Robinson, S. C. (2020). Trust, transparency, and openness: How inclusion of cultural values shapes Nordic national public policy strategies for artificial intelligence (AI). *Technology in Society*, 63, 101421. <https://doi.org/10.1016/j.techsoc.2020.101421>
- Rogers, E. M. (2003). *Diffusion of innovations* (5th ed.). Free Press.
- Ryan, M. (2020). In AI we trust: Ethics, artificial intelligence, and reliability. *Science and Engineering Ethics*, 26(5), 2749–2767. <https://doi.org/10.1007/s11948-020-00228-y>
- Ryan, M. (2022). The social and ethical impacts of artificial intelligence in agriculture: mapping the agricultural AI literature. *AI and Society*, 1–13. <https://doi.org/10.1007/s00146-021-01377-9>
- Saßmannshausen, T., Burggräf, P., Wagner, J., Hassenzahl, M., Heupel, T., & Steinberg, F. (2021). Trust in artificial intelligence within production management—an exploration of antecedents. *Ergonomics*, 1–18. <https://doi.org/10.1080/00140139.2021.1909755>
- Sanclemente, G. L. (2021). Reliability: Understanding cognitive human bias in artificial intelligence for national security and intelligence analysis. *Security Journal*, 1–21. <https://doi.org/10.1057/s41284-021-00321-2>
- Schmidt, P., Biessmann, F., & Teubner, T. (2020). Transparency and trust in artificial intelligence systems. *Journal of Decision Systems*, 29(4), 260–278. <https://doi.org/10.1080/12460125.2020.1819094>
- Shao, Z., Li, X., Guo, Y., & Zhang, L. (2020). Influence of service quality in sharing economy: Understanding customers' continuance intention of bicycle sharing. *Electronic Commerce Research and Applications*, 40, 100944.
- Sharan, N. N., & Romano, D. M. (2020). The effects of personality and locus of control on trust in humans versus artificial intelligence. *Heliyon*, 6(8), e04572. <https://doi.org/10.1016/j.heliyon.2020.e04572>
- Sheridan, T. B. (2019). Individual differences in attributes of trust in automation: Measurement and application to system design. *Frontiers in Psychology*, 10, 1117. <https://doi.org/10.3389/fpsyg.2019.01117>
- Shi, S., Gong, Y., & Gursay, D. (2021). Antecedents of trust and adoption intention toward artificially intelligent recommendation systems in travel planning: A heuristic-systematic model. *Journal of Travel Research*, 60(8), 1714–1734. <https://doi.org/10.1177/0047287520966395>
- Shin, D. (2021). The effects of explainability and causability on perception, trust, and acceptance: Implications for explainable AI.

- International Journal of Human-Computer Studies*, 146, 102551. <https://doi.org/10.1016/j.ijhcs.2020.102551>
- Skulimowski, A. M. (2013). Universal intelligence, creativity, and trust in emerging global expert systems. *International Conference on Artificial Intelligence and Soft Computing*. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-642-38610-7_53.
- Söllner, M., Hoffmann, A., & Leimeister, J. M. (2016). Why different trust relationships matter for information systems users. *European Journal of Information Systems*, 25(3), 274–287. <https://doi.org/10.1057/ejis.2015.17>
- Starke, G., van den Brule, R., Elger, B. S., & Haselager, P. (2021). Intentional machines: A defence of trust in medical artificial intelligence. *Bioethics*, 36(2), 154–161. <https://doi.org/10.1111/bioe.12891>
- Stumpf, S., Rajaram, V., Li, L., Burnett, M., Dietterich, T., Sullivan, E., & Herlocker, J. (2007). Toward harnessing user feedback for machine learning. In *Proceedings of the 12th international Conference on Intelligent User Interfaces, Honolulu Hawaii, USA*. <https://doi.org/10.1145/1216295.1216316>
- Sullivan, Y., de Bourmont, M., & Dunaway, M. (2020). Appraisals of harms and injustice trigger an eerie feeling that decreases trust in artificial intelligence systems. *Annals of Operations Research*, 1–24. <https://doi.org/10.1007/s10479-020-03702-9>
- Sutrop, M. (2019). Should we trust artificial intelligence? *Trames*, 23(4), 499–522. <https://doi.org/10.3176/tr.2019.4.07>
- Thiebes, S., Lins, S., & Sunyaev, A. (2021). Trustworthy artificial intelligence. *Electronic Markets*, 31(2), 447–464. <https://doi.org/10.1007/s12525-020-00441-4>
- Tornatzky, L. G., Fleischer, M., & Chakrabarti, A. K. (1990). *Processes of technological innovation*. Lexington Books.
- Triberti, S., Durosini, I., Curigliano, G., & Pravettoni, G. (2020). Is explanation a marketing problem? The quest for trust in artificial intelligence and two conflicting solutions. *Public Health Genomics*, 23(1–2), 2–5. <https://doi.org/10.1159/000506014>
- Troshani, I., Rao Hill, S., Sherman, C., & Arthur, D. (2020). Do we trust in AI? Role of anthropomorphism and intelligence. *Journal of Computer Information Systems*, 1–11. <https://doi.org/10.1080/08874417.2020.1788473>
- Tuteja, G., Gupta, S., & Garg, V. (2016). Consumer trust in internet shopping: An empirical investigation. *Paradigm*, 20(2), 191–215. <https://doi.org/10.1177/0971890716670723>
- Vasiljeva, T., Kreituss, I., & Lulle, I. (2021). Artificial intelligence: The attitude of the public and representatives of various industries. *Journal of Risk and Financial Management*, 14(8), 339. <https://doi.org/10.3390/jrfm14080339>
- Vassilikopoulou, A., Lepetos, A., & Siomkos, G. (2018). Crises through the consumer lens: The role of trust, blame and risk. *Journal of Consumer Marketing*, 35(5), 502–511. <https://doi.org/10.1108/JCM-02-2016-1721>
- Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly*, 425–478. <https://doi.org/10.2307/30036540>
- Vourgidis, I., Mafuma, S. J., Wilson, P., Carter, J., & Cosma, G. (2018). Medical expert systems - a study of trust and acceptance by healthcare stakeholders. *Proceedings of the UK Workshop on Computational Intelligence*. https://doi.org/10.1007/978-3-319-97982-3_9
- Weber-Lewerenz, B. (2021). Corporate digital responsibility (CDR) in construction engineering—Ethical guidelines for the application of digital transformation and artificial intelligence (AI) in user practice. *SN Applied Sciences*, 3(10), 1–25. <https://doi.org/10.1007/s42452-021-04776-1>
- Webster, J., & Watson, R. T. (2002). Analyzing the past to prepare for the future: Writing a literature review. *MIS Quarterly*, 26(2), xiii–xxiii.
- Wei, K., Li, Y., Zha, Y., & Ma, J. (2019). Trust, risk and transaction intention in consumer-to-consumer e-marketplaces: An empirical comparison between buyers' and sellers' perspectives. *Industrial Management & Data Systems*, 119(2), 331–350. <https://doi.org/10.1108/IMDS-10-2017-0489>
- Weitz, K., Schiller, D., Schlagowski, R., Huber, T., & André, E. (2019). Do you trust me? Increasing user-trust by integrating virtual agents in explainable AI interaction design. *Proceedings of the 19th ACM International Conference on Intelligent Virtual Agents*. <https://doi.org/10.1145/3308532.3329441>.
- Wibowo, S., and Mubarak, S. (2020). Exploring stakeholders perceived risk and trust towards their intention to adopt cloud computing: A theoretical framework. *Proceedings of the Pacific Asia Conference on Information Systems*. <https://aisel.aisnet.org/pacis2020/1/>
- Wibowo, S., Grandhi, S., Duan, S., and Mubarak, S. (2020). Trust-building framework for blockchain technology adoption: The moderating role of quality practices. *Proceedings of the Australasian Conference on Information Systems*. <https://aisel.aisnet.org/acis2020/90/>
- Wilkes, R. (2014). Trust in government: A micro–macro approach. *Journal of Trust Research*, 4(2), 113–131. <https://doi.org/10.1080/21515581.2014.889835>
- Winfield, A. F., & Jirotk, M. (2018). Ethical governance is essential to building trust in robotics and artificial intelligence systems. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 376(2133), 20180085. <https://doi.org/10.1098/rsta.2018.0085>
- Wirtz, B. W., Weyerer, J. C., & Geyer, C. (2019). Artificial intelligence and the public sector - applications and challenges. *International Journal of Public Administration*, 42(7), 596–615. <https://doi.org/10.1080/01900692.2018.1498103>
- Wolfswinkel, J. F., Furtmueller, E., & Wilderom, C. P. M. (2013). Using grounded theory as a method for rigorously reviewing literature. *European Journal of Information Systems*, 22(1), 45–55. <https://doi.org/10.1057/ejis.2011.51>
- Xin, H., Techatassanasoontorn, A. A., & Tan, F. B. (2015). Antecedents of consumer trust in mobile payment adoption. *Journal of Computer Information Systems*, 55(4), 1–10. <https://doi.org/10.1080/08874417.2015.11645781>
- Yang, R., & Wibowo, S. (2020). Risks and uncertainties in citizens' trust and adoption of e-government: A proposed framework. *ACIS 2020 Proceedings*, 80, <https://aisel.aisnet.org/acis2020/80>
- Yang, R., & Wibowo, S. (2022). The effects of risk and uncertainty perceptions on tourists' intention to visit former epicenters of COVID-19 post-pandemic: Evidence from China. *Leisure Sciences*, 1–24. <https://doi.org/10.1080/01490400.2022.2061653>
- Yang, J., & Zhang, Y. (2020). Research on the factors affecting consumer trust in green residences - based on SEM model and SPSS data processing software. *The International Journal of Electrical Engineering & Education*, 0020720920930351. <https://doi.org/10.1177/0020720920930351>
- Yang, Q., Pang, C., Liu, L., Yen, D. C., & Tarn, J. M. (2015). Exploring consumer perceived risk and trust for online payments: An empirical study in China's younger generation. *Computers in Human Behavior*, 50, 9–24. <https://doi.org/10.1016/j.chb.2015.03.058>
- Yang, Z., Van Ngo, Q., Chen, Y., Nguyen, C. X. T., & Hoang, H. T. (2019). Does ethics perception foster consumer repurchase intention? Role of trust, perceived uncertainty, and shopping

- habit. *SAGE Open*, 9(2), 2158244019848844. <https://doi.org/10.1177/2158244019848844>
- Yen, C., & Chiang, M. C. (2021). Trust me, if you can: A study on the factors that influence consumers' purchase intention triggered by chatbots based on brain image evidence and self-reported assessments. *Behaviour & Information Technology*, 40(11), 1177–1194. <https://doi.org/10.1080/0144929X.2020.1743362>
- Zhang, S., Meng, Z., Chen, B., Yang, X., & Zhao, X. (2021). Motivation, social emotion, and the acceptance of artificial intelligence virtual assistants -trust-based mediating effects. *Frontiers in Psychology*, 3441. <https://doi.org/10.3389/fpsyg.2021.728495>
- Zhu, J., & Liu, W. (2020). A tale of two databases: The use of web of science and Scopus in academic papers. *Scientometrics*, 123, 321–335. <https://doi.org/10.1007/s11192-020-03387-8>
- Publisher's note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.
- Springer Nature or its licensor holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.